

I. Axes Cut List

Axes + Bed Holder + Spool Holder:

- [12] Rods - 13- $\frac{7}{8}$ " long
- [1] Rod - Spool Holder - 15.5"
- [2] Rods - Spool Holder - 10"
- [5] Belt - 31 inch

II. Introduction

The axis assemblies act as rails that move the extruder and print bed in three-dimensional space to allow the printing of 3D objects. The axis assemblies are mounted on the frame. The Y axes and Z axes mount directly to the frame, and the X axis mounts between the Y axes.

Note that this manual repeats a lot of the content between the axes. This is intentional - so that builders can look at the complete instructions without having to skip back between pages. This is useful especially when multiple teams are working on multiple axes at the same time. **Do not deviate from the procedures shown thinking that you know better - as there are slight variations between axes.** You will have to take the axes apart if you don't follow the directions exactly.

III. Functional Knowledge

The Axis Assemblies include a motor system that moves the print head with great precision. It is run by a stepper motor, which is a type of motor that can rotate in very small "steps," to control exactly how much distance the print head travels.

This motor system is connected to a "belt drive", where a belt is wrapped around the shaft of the motor, so that when the motor turns, the belt moves linearly and therefore moves the print-head linearly.

There are five Axis Assemblies in total: the x-axis, y-right axis, y-left axis, z-front axis, and z-back axis. The x-axis' function is to move the print head left and right. The two y-axes' function is to move the print head forward and backward. The two z-axes' function is to move the print bed up and down.

An advantage of the design you will build today is that it is modular. The left y-axis and Z-Back axis are exactly the same; similarly, the right y-axis and Z-Front axis are exactly the same. So to construct the five-piece axis system, you only have to learn how to build three unique parts - and of the 3 unique they are all almost identical.

IV. Instructional

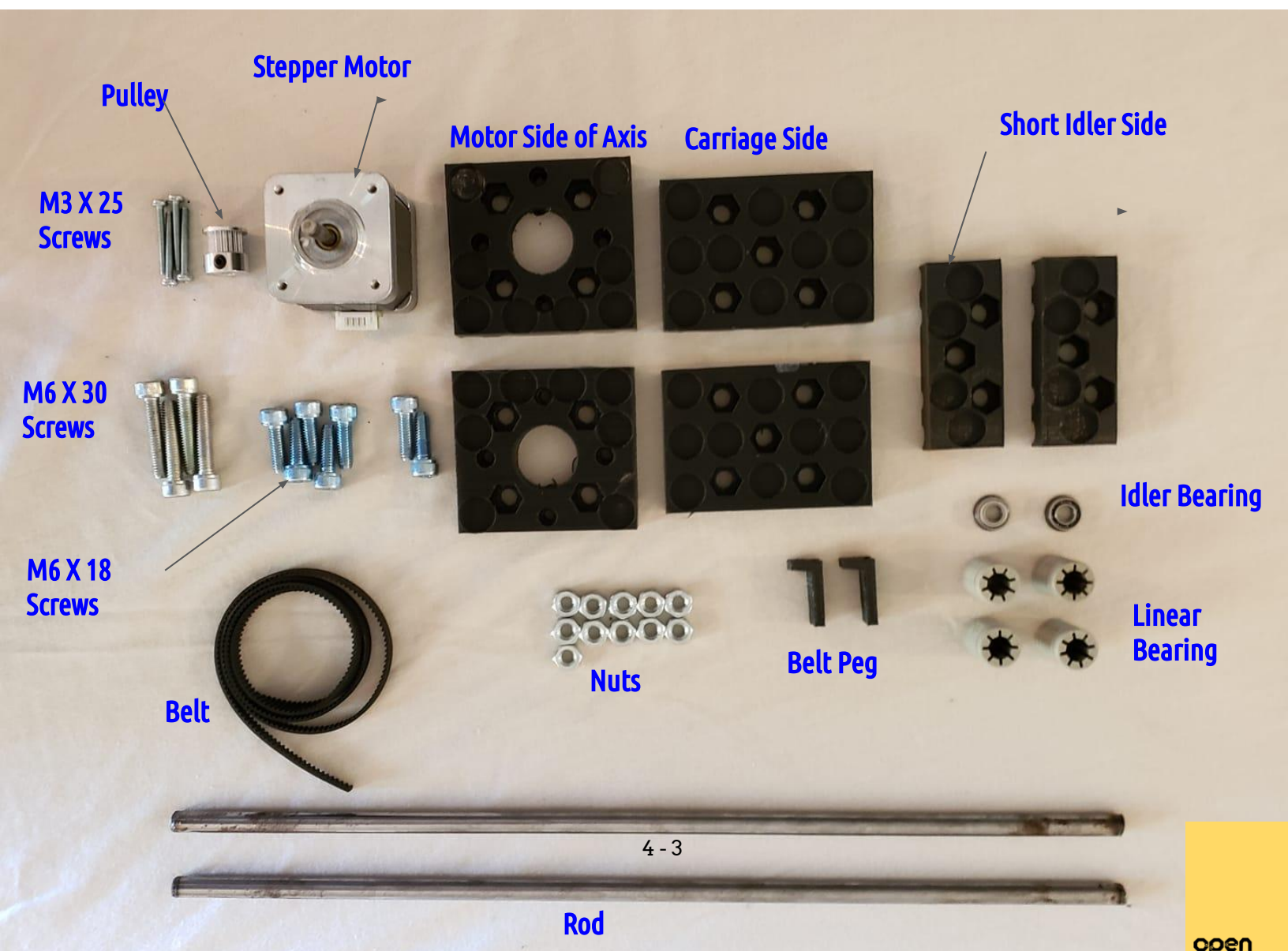
Axis Assemblies

Y-Left/Z-Back Axis - Parts

Two axes in the printer are identical - the Y-Left and Z-Back axes.

Note that the instructions for the Y-left and Z-front axis are meant to be followed twice: once to build the y-left axis, and again to build the z-front axis.

You can choose to make both of these axes at the same time, or you can make them one at a time.



III. Instructional

Y-Left/Z-Back Axis

Tools & Supplies

- Power Drill
- Box Cutter or Razor Blade
- Allen Key -- Metric 5
- Allen Key -- Metric 2 - for pulleys
- Needle nose Pliers
- Piece of Sandpaper - 220 grit or fine grit
- Phillips Screwdriver
- Spray of oil
- Scissors for electrical tape
- 3 mm + 6 mm drill bits to clean up the plastic prints if there are imperfections

We recommend 2 people to build an axis. Have an extra power drill and screwdriver for each additional person building.

III. Instructional - Preparation

Y-Left/Z-Back Axis

Step 1.1: Setup & Prepare - Get the correct pieces out.

Step 1.2: Clean up carriage half-pieces, idler half pieces, and motor holder half pieces of big and small holes

Use a box cutter or razor blade to clean out larger curves so that there's no extra plastic covering the holes.

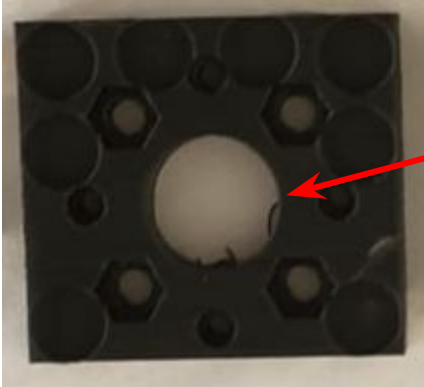
You can also use a cordless power drill with a drill bit to poke through holes and ream the sides of the holes. This means you put the drill bit through a hole, turn on the power drill, and applying pressure to the sides of the hole so that the twisting of the drill whittles away the sides of the hole.

Use a drill bit to clean out holes. Hole should be smooth.

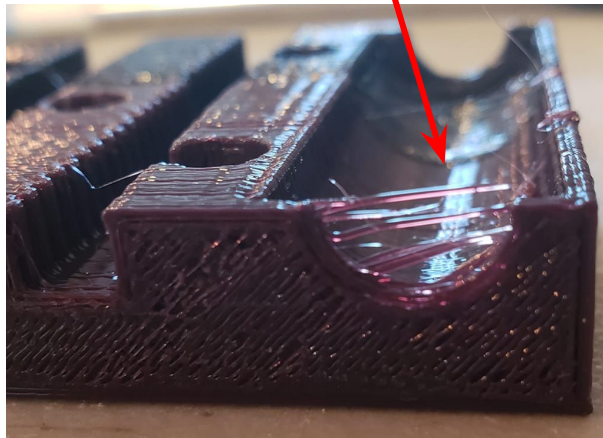
This ensures that rods can be inserted and held in the plastic pieces, and that belts do not rub against the plastic pieces .

III. Instructional - Preparation

Y-Left/Z-Back Axis



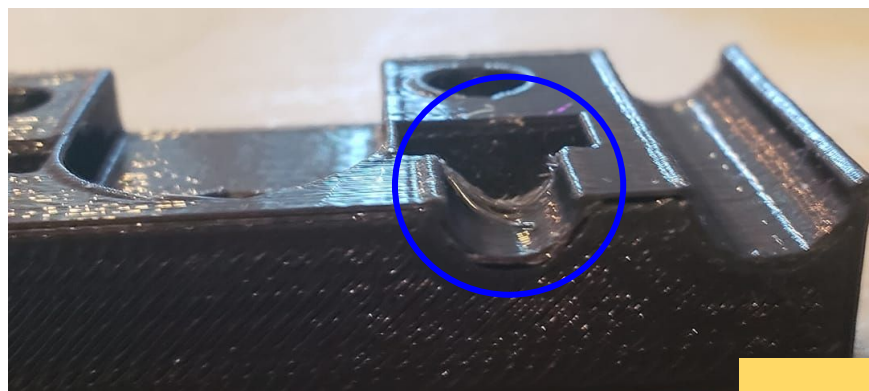
Clean this up



Carriage half piece with excess plastic threading



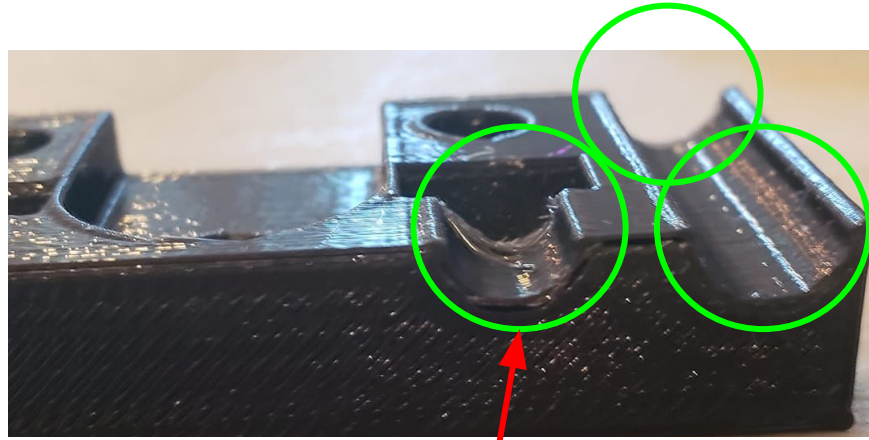
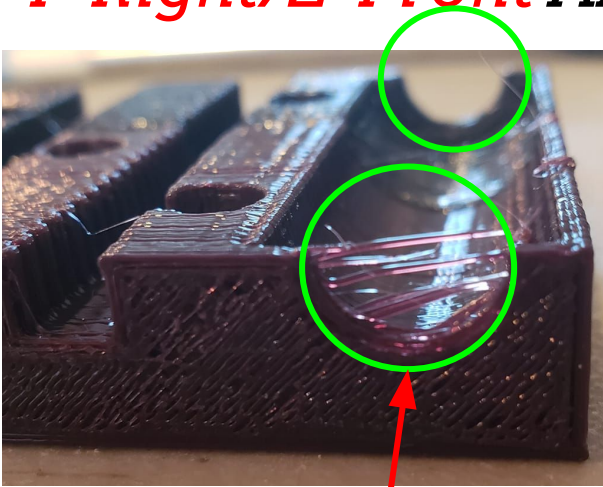
Motor Holder Half Piece with excess plastic



Motor Holder Half Piece cleaned

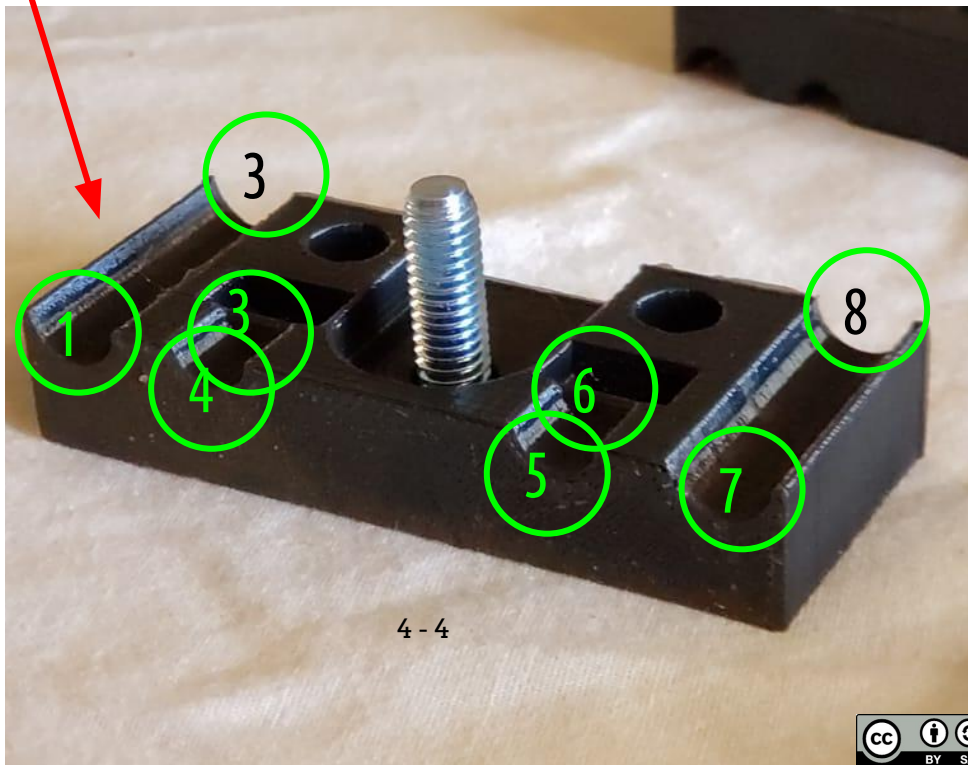
III. Instructional - Preparation

Y-Right/Z-Front Axis



Step 1.3: Plastic Parts Quality Control

- ☐ Each Carriage Half Piece has 4 clean semi-circles
- ☐ Each motor holder piece has 6 clean semi-circles
- ☐ Each idler piece 8 clean semi-circles
- ☐ Motor piece has 4 clear holes for the 6 mm bolts
- ☐ Motro piece has 4 clear holes for the 3 mm screws
- ☐ Carriage piece has 4 clear holes (mid one is not used)
- ☐ End piece has 3 clear holes



4 - 4

III. Instructional

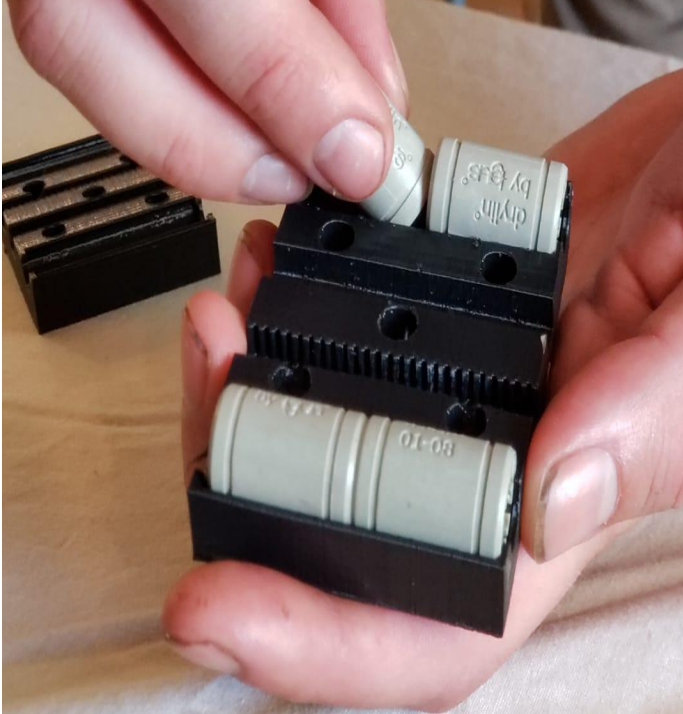
Carriage > Y-Left/Z-Back Axis

Step 1.4: Test bearings fit

Place the four bearings in one side of the Axis carriage. Add another side of carriage on top and hold them together.

It needs to be tight enough for a good print, but loose enough for the carriage to slide on the rods.

Follow the next step to test that the bearings fit well.



III. Instructional

Carriage > Y-Left/Z-Back Axis

Test Method 1: Carriage Shake

This test ensures that the bearing is tight enough.

Put bearings in carriage. Close the carriage with your hand and compress the carriage piece tightly. Shake it. If the bearings shake inside the carriage, you will need to adjust the bearing size so that it is a snug fit inside the carriage and doesn't shake.



4 - 4

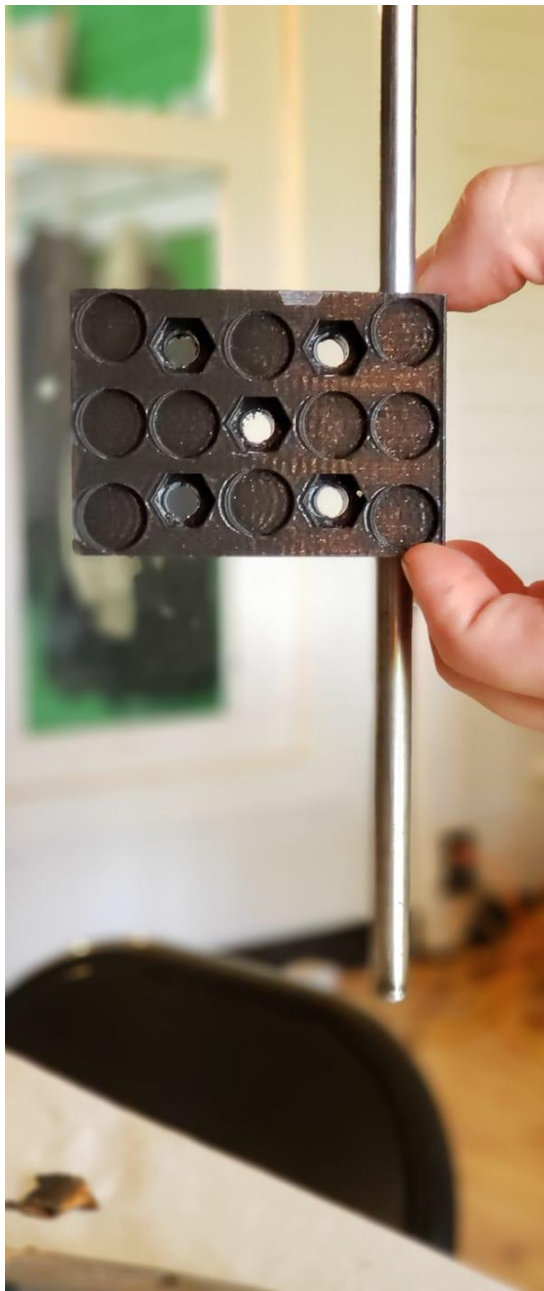
III. Instructional

Carriage > Y-Left/Z-Back Axis

Test Method 2: Rod Slide

This test ensures that the bearing is not too tight.

Put the rod through the carriage. Turn the carriage so that the rod is vertical. Rod should slide or drop through as you hold the carriage.



Motors will be pulling a belt attached to this carriage with 20 pounds of force. So the rod shouldn't just slip through the carriage super fast; instead, **there should be a little resistance**. It's good if you have to shake the carriage for the rod to slide through.

If the rod slides too quickly through the carriage without any resistance, you should probably adjust the bearings by making them tighter.

It's likely the rod won't be adjusted exactly right. See next step to use tape to adjust the tightness.

III. Instructional

Carriage > Y-Left/Z-Back Axis

Step 1.4: Adjust bearings if needed

Wrap each bearing in one layer of electrical tape. Tape so that it just barely overlaps with itself. We recommend electrical tape because it's on hand in this build, but any tape will work.

Repeat as necessary.

Why does tape make it tighter?

If the rod slips through really quickly, then adding tape to the bearings will give it more resistance, even though the rod is always slipping through the same place.

Why can tape make it too tight?

Too much tape will compress the bearing and keep it from sliding.

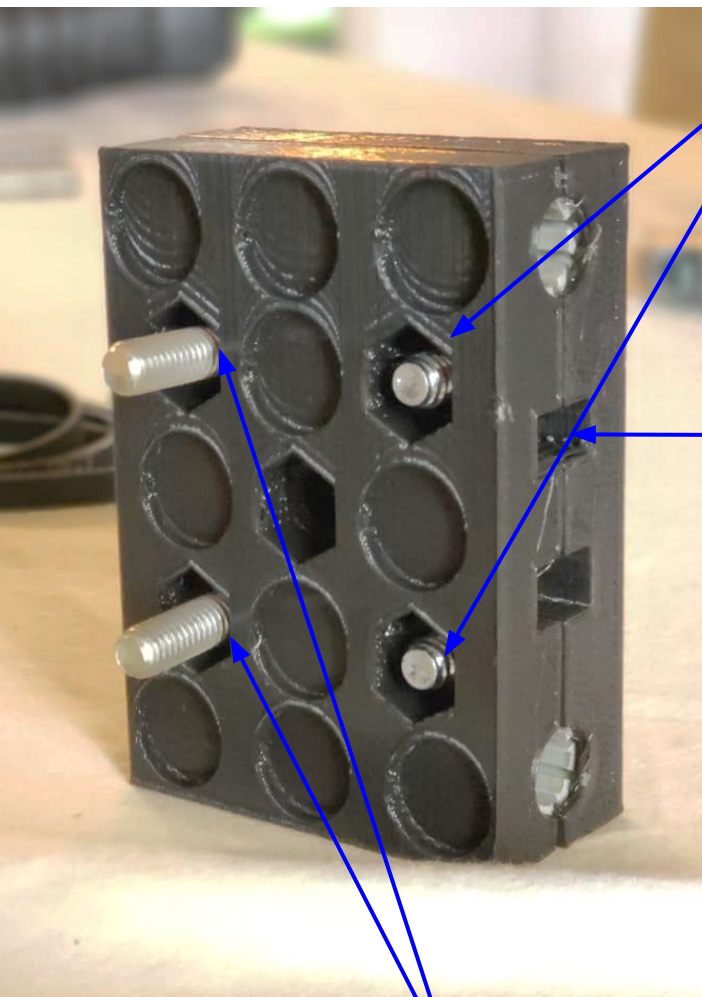


III. Instructional

Carriage > Y-Left/Z-Back Axis

Step 2.1: Assemble Carriage

Four bolts will be used to secure the carriage piece as shown. Note the proper alignment of the rectangular belt holes.



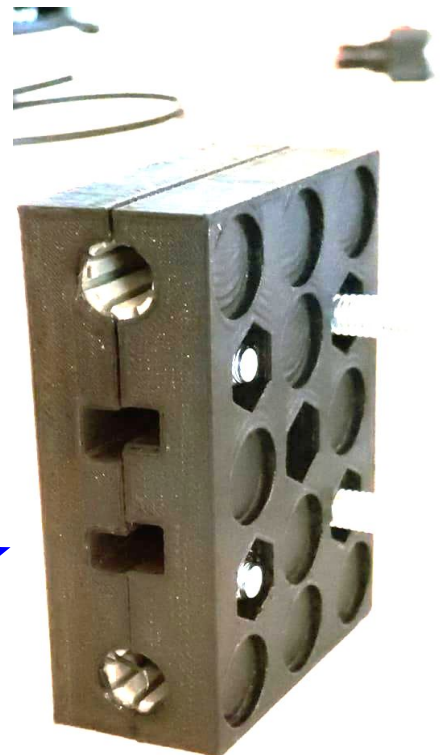
M6 X 18 bolts

Note there are 2 different belt holes. One is small and ribbed. The other is larger and smooth. The belt peg goes into the small, ribbed hole.

Correct alignment of holes

M6 X 30 bolts

Incorrect alignment of holes mismatches the small and large holes 4 - 4



III. Instructional

Carriage > Y-Left/Z-Back Axis

Step 2.2: Tighten bolts into the nuts

Use an electric drill on the lowest torque setting to secure nuts to the four bolts.



You've completed an axis carriage.

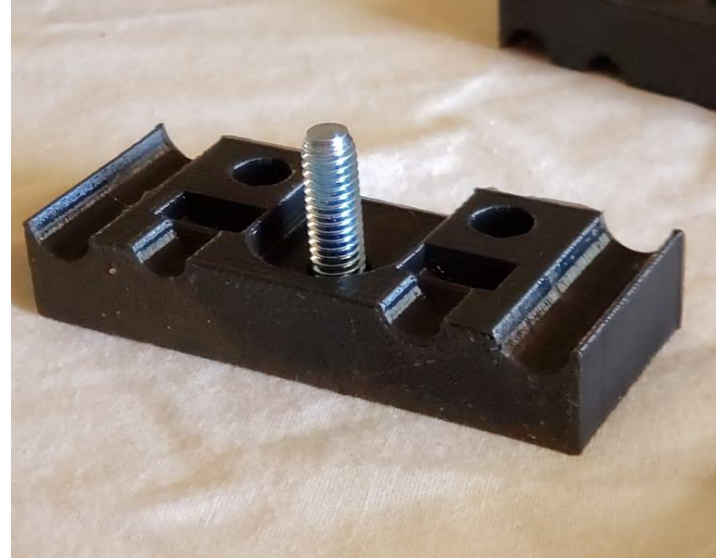
Optional: Repeat the Carriage Assembly three more times so you have a total of four built for all the Axis Assemblies.

III. Instructional

Idler > Y-Left/Z-Back Axis

Step 3.1: Insert M6x18 Bolt

Begin assembling the idler by placing a M6x18 bolt through one side.



Step 3.2: Insert two bearing pieces

Insert the bearings one on top of the other on the bolt. Flanges should be facing outside, as shown.



Step 3.3: Identify how to put the idler piece together correctly

Make sure the idler pieces align correctly. These circular holes for the rods should line up with each other.

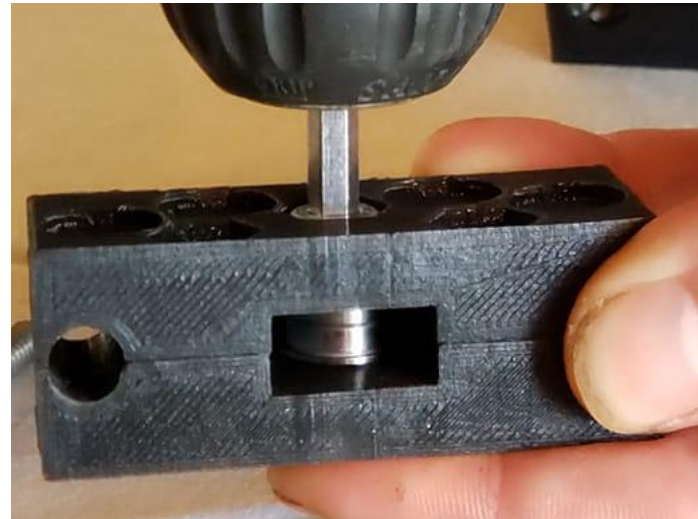


III. Instructional

Idler > Y-Left/Z-Back Axis

Step 3.4: Secure bearing with one M6x18 bolt

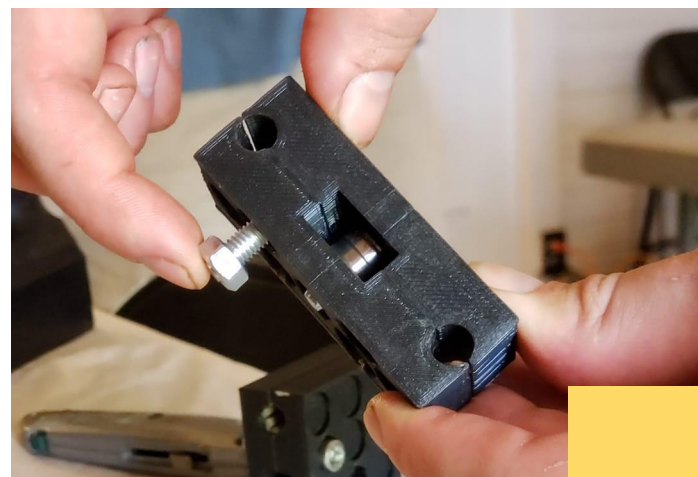
Put a nut on the back of the bolt and secure fully with an electrical drill on the lowest setting.



Step 3.5: Put 1 m6x30 screw into correct hole

With idler piece laying flat in front of you, make sure the bearing hole is pointing away from you. Insert one M6x30 screw into the top right hole in the same direction as the existing screw. Secure loosely with a nut on the opposite side -- because you will be attaching this to the frame later.

Note: what you see in this picture is the nut, NOT the bolt head.

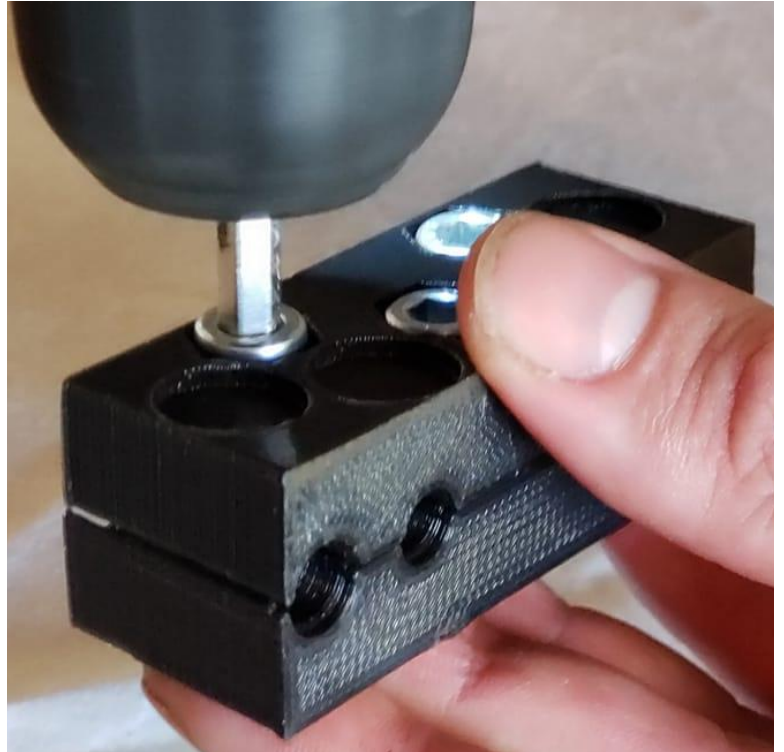


III. Instructional

Idler > Y-Left/Z-Back Axis

Step 3.6: Secure a M6x18 bolt through the last hole using a nut and the electric drill.

Screw the nut on, not so that it's tightened, but kept very loose as shown. We will loosen this piece later on to attach to the frame.



Your idler is ready to go. Put to the side and continue on to assembling the motor.



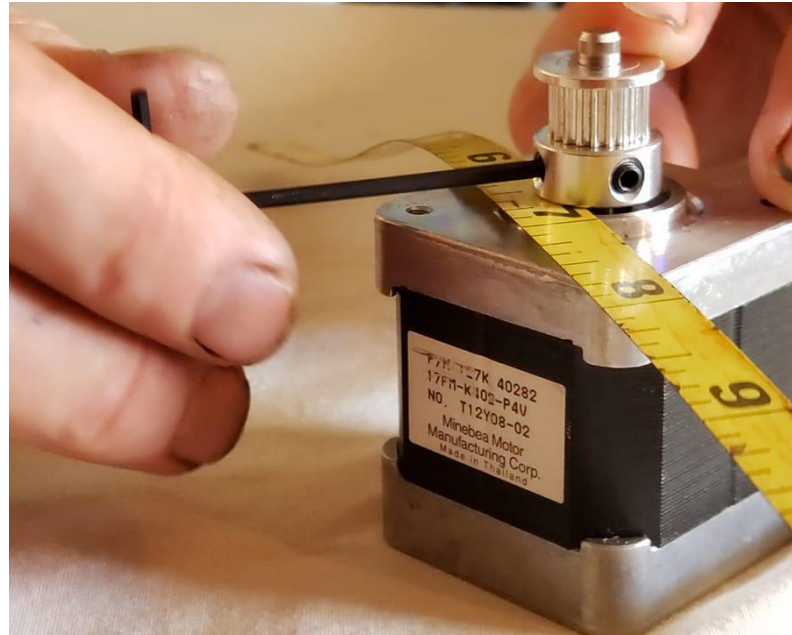
III. Instructional

Motor > Y-Left/Z-Back Axis

Step 4.1: Put pulley on motor

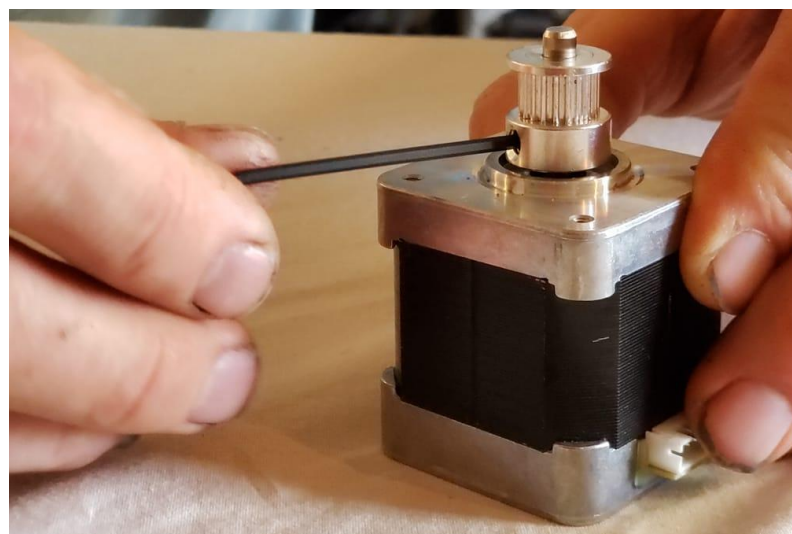
Place the pulley on the motor using a thin spacer (provided) that the pulley rests upon. Tighten the first set screw with an Allen wrench.

The space prevents the pulley from rubbing on the motor housing, and assists in aligning the pulley gear to the belt channel.



Step 4.2: Tighten pulley set screw onto motor shaft

With the first set screw tightened, you can remove the spacer. Then tighten the second set screw.



III. Instructional

Motor > Y-Left/Z-Back Axis

Step 4.3: **Quality Control Check**

Flip the Allen wrench and tighten down the set screw as far as you can, pressing down on the wrench near its midpoint for greatest pressure. When you hear the set screw "pop.", you can be sure you've set it as tightly as possible.

The pop sound occurs when the small screw head slips along the motor shaft. It doesn't get any tighter than that. This tight fit ensures the belt will be pulled effectively by the motor shaft.

It's also OK if you don't hear a pop sound. As long as you've tightened the screw as hard as you can, then it should be fine.

In the next steps you will be assembling the motor pieces.



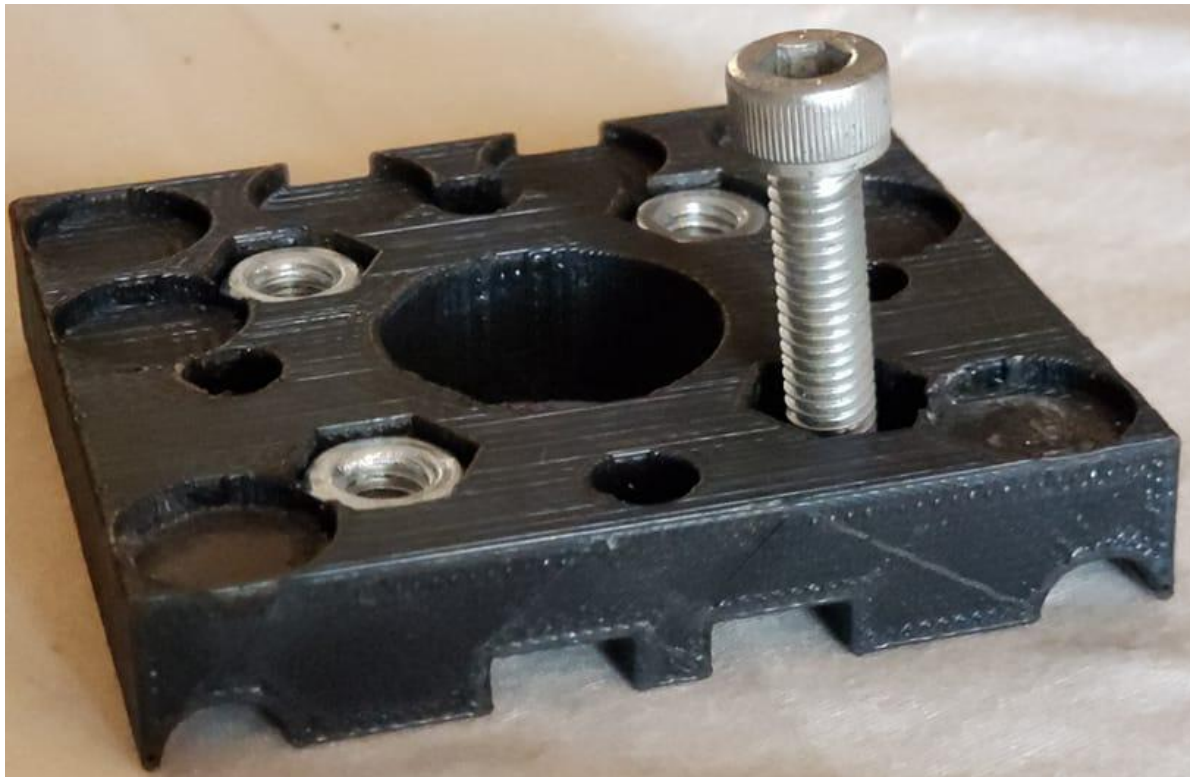
III. Instructional

Motor > Y-Left/Z-Back Axis

Step 5.1: Setup Motor half-piece with three nuts, one M6x30 bolt

Note the position of the rectangular belt holes. Insert one M6x30 bolt and three nuts as shown.

When you put the motor on top of the motor half-piece, this lets you access the screws because the motor will be covering the pieces later on. This will make sense in pictures down the line.

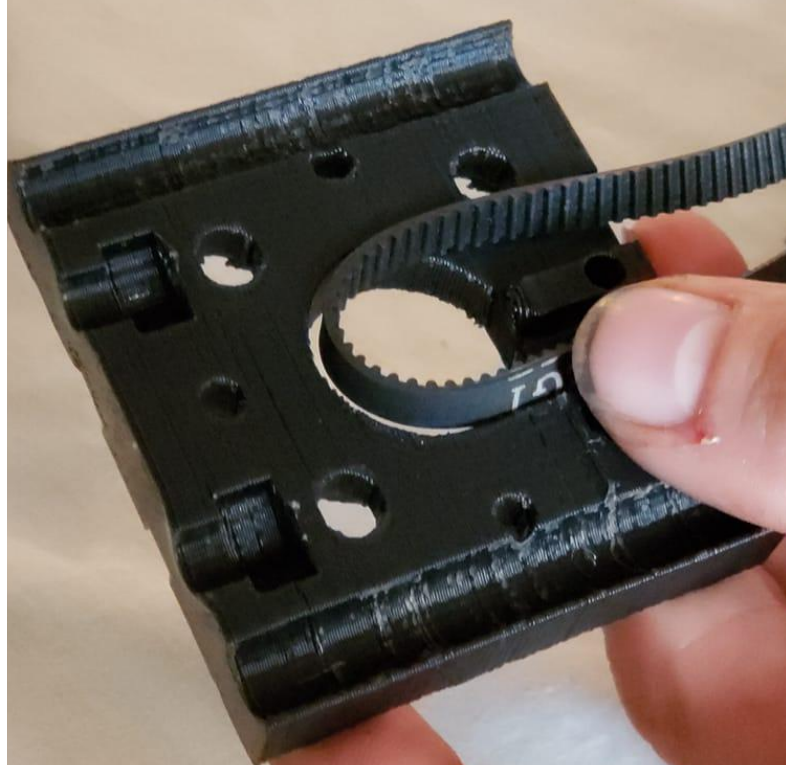


III. Instructional

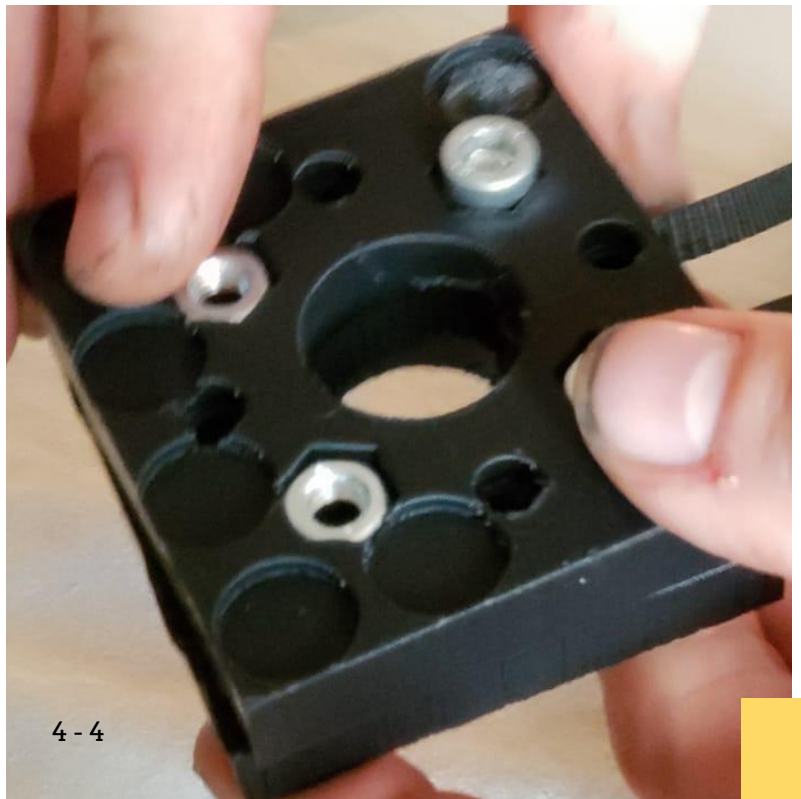
Motor > Y-Left/Z-Back Axis

Step 5.2: Set up the other half piece with the belt set inside.

There's a nice channel for the belt on the other half piece. Insert the belt with the toothed side facing inwards.



Step 5.3: Close up the two half pieces



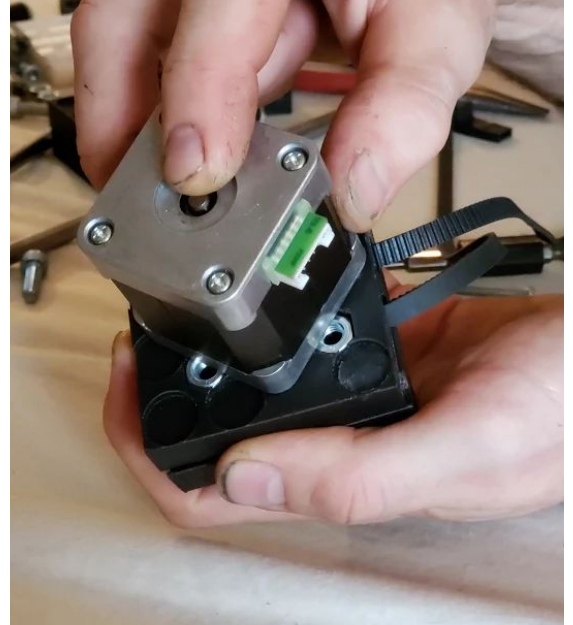
4 - 4

III. Instructional

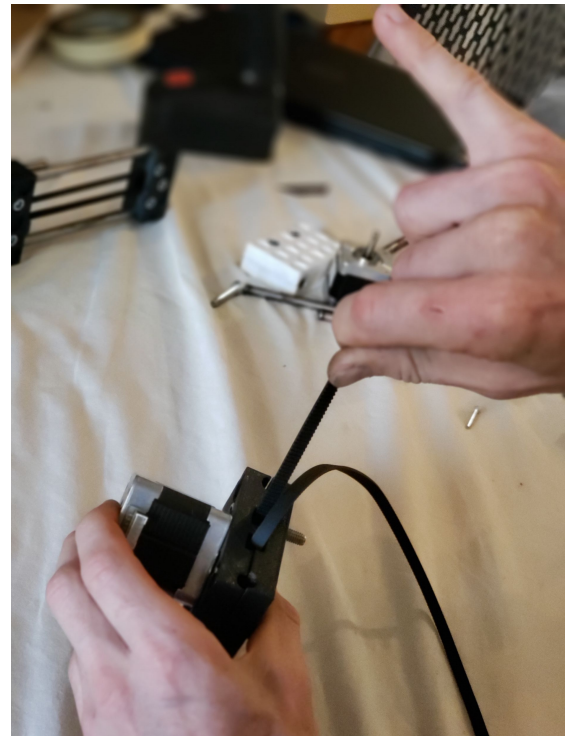
Motor > Y-Left/Z-Back Axis

Step 5.3: Test out the pulley

Insert the motor piece such that the cable plug faces towards the bottom right. Hold it tight with your hand for now.



Make sure the belt is moving along the motor pulley by giving it a gentle pull.



III. Instructional

Motor > Y-Left/Z-Back Axis

Step 5.4: Attach the step motor

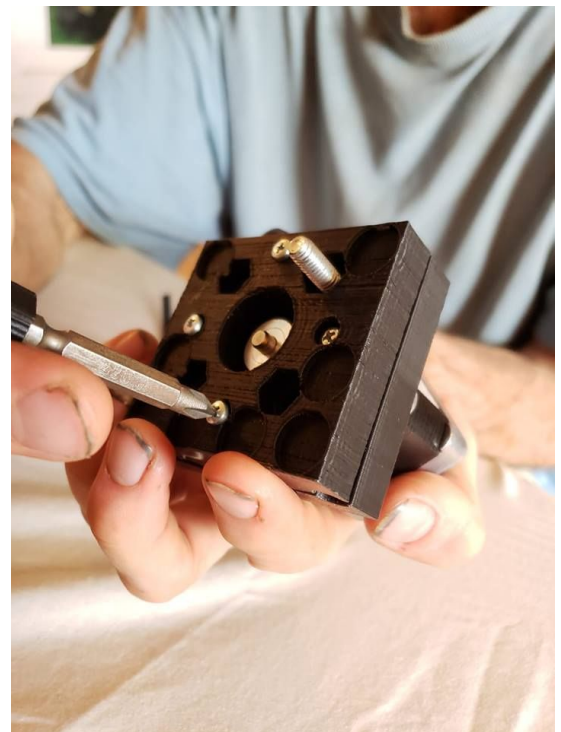
Insert the first motor screw. Tighten it down 3/4 with a screwdriver. Leave it a little loose so it has some flexibility. It will be fully tightened later.



Insert the three remaining screws.

Tighten each so they remain somewhat loose.

Continued steps on the next page.



III. Instructional

Motor > Y-Left/Z-Back Axis

Step 5.4: Attach the step motor (continued)

Insert M6x18 bolts into the three remaining holes.



Tighten each into the nut using an electric drill.



III. Instructional

Motor > Y-Left/Z-Back Axis

Step 5.5: Test the motor assembly

Now you want to make sure everything is secured properly. Perform the dangle test - complete with extended pinky for extra class.



III. Instructional

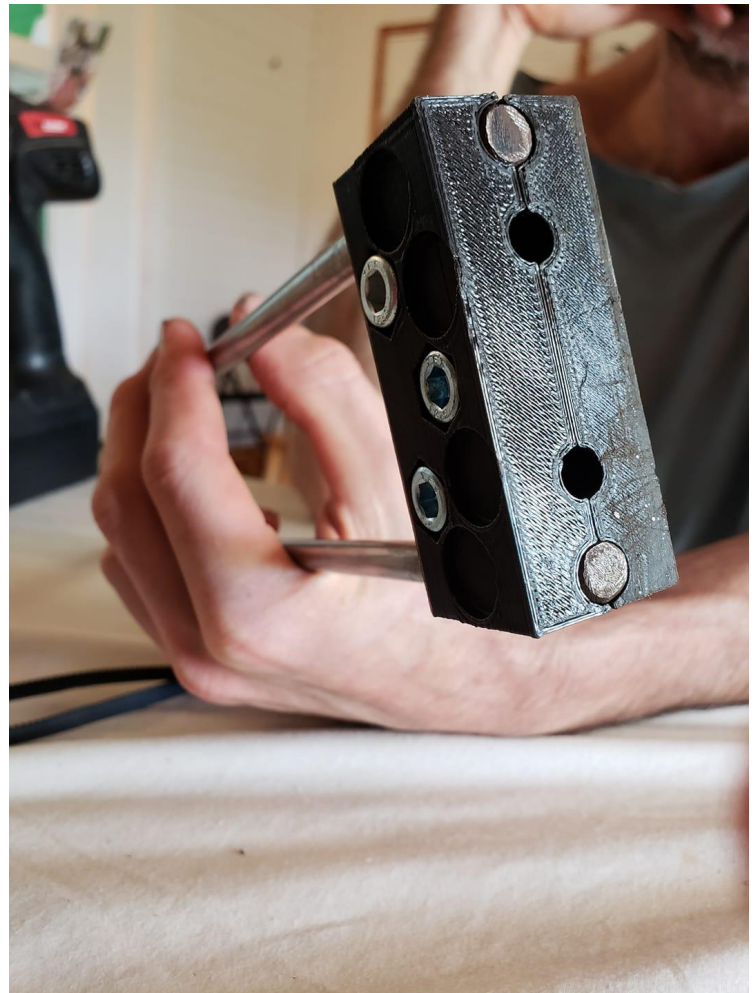
Axis Assembly > Y-Left/Z-Back Axis

Step 6.1 : Place the idler on the rods

Now we'll put it all together. Insert the rods into the completed idler piece.

If it is difficult to fit the rods on, just loosen the bolts until the rods slide in. Then re-tighten.

Make sure the rods are flush against the bottom.



III. Instructional

Axis Assembly > Y-Left/Z-Back Axis

Step 6.2 : Tighten the idler bolts

Use an electric drill and a nut to tighten the two shorter bolts on the idler. This will tighten the idler around the rods

Leave the longer bolt loose. This will be used to attach the Axis Assembly to the frame later.



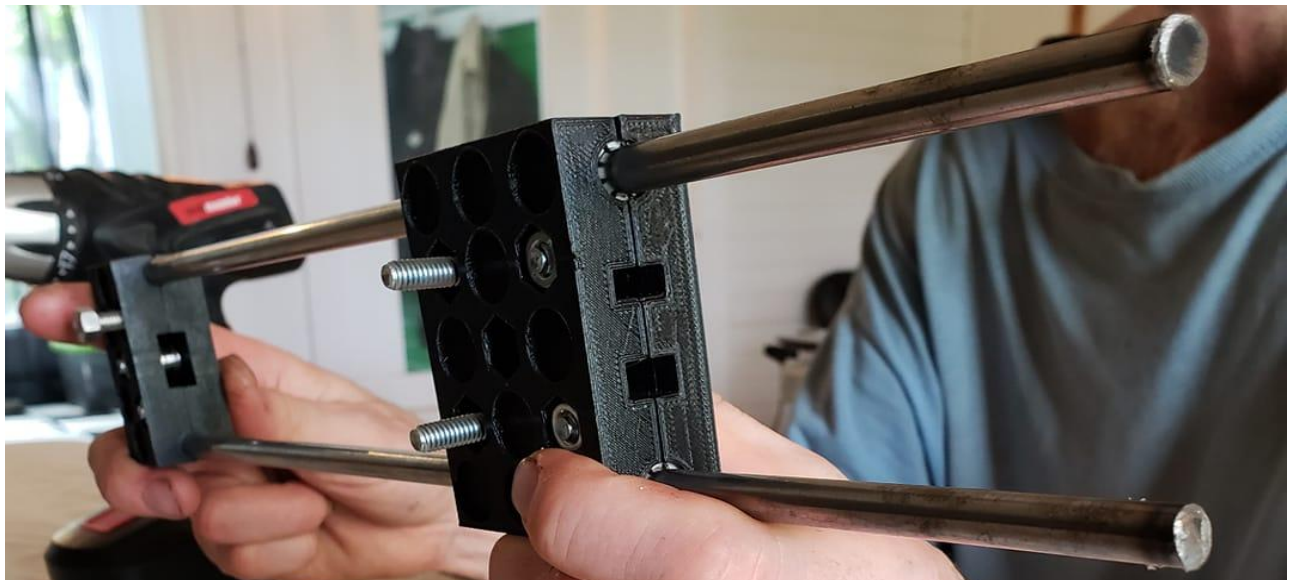
III. Instructional

Axis Assembly > Y-Left/Z-Back Axis

Step 6.3: Put carriage assembly onto rods

Slide the completed carriage piece onto the rods with the long bolt ends closer to the idler as shown.

This design allows the carriage to travel farther and therefore increases the printing space.



III. Instructional

Axis Assembly > Y-Left/Z-Back Axis

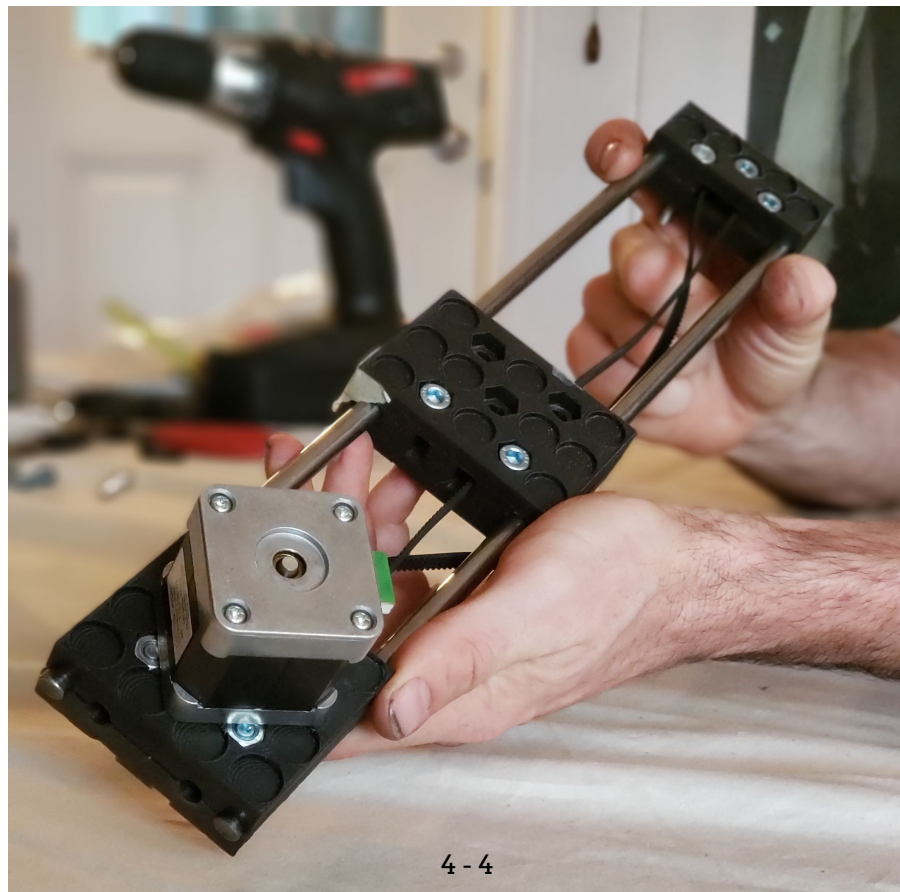
Step 6.4: Put motor piece onto rods

Push the motor piece onto the rods so the motor is facing the carriage bolt heads as shown.

If it is difficult to fit the rods on, just loosen the bolts until the rods slide in. Then re-tighten.

Step 6.5: **Quality Control Check**

Check the idler that the rods are still flush (see image on previous page).



4 - 4

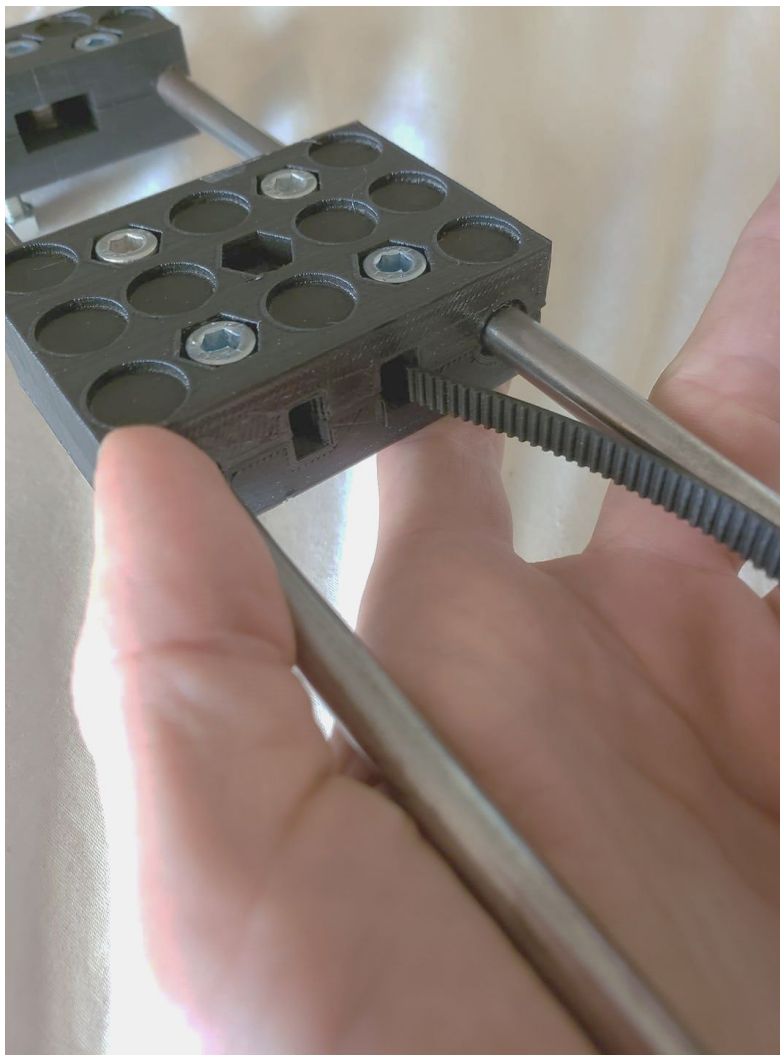
III. Instructional

Belt Threading > Y-Left/Z-Back Axis

Step 7.1: Thread belt into carriage

You are going to tighten the belt through the motor piece, carriage, and idler. Before moving on, take a close look at the carriage piece. Notice it has two belt holes, one of which is larger than the other.

Insert one end of the belt **through the larger belt hole** on the carriage piece.



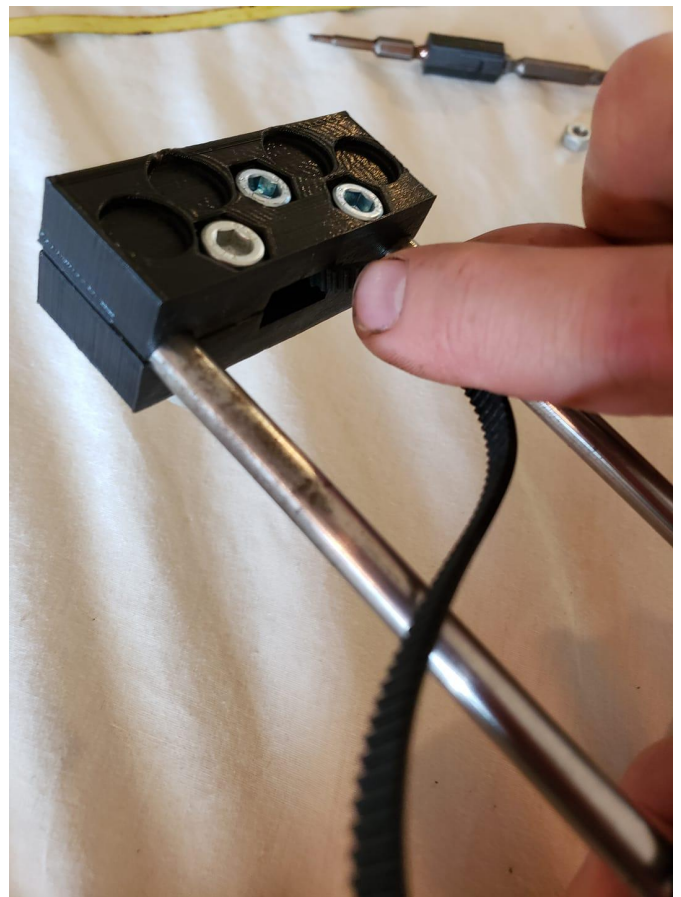
III. Instructional

Belt Threading > Y-Left/Z-Back Axis

Step 7.2: Thread belt around idler bearing

Continue threading the belt around the bearing inside of the idler.

Pro Tip. Give a slight curve in the right direction to your belt before sliding it inside the idler. It will tend to get around the bearing more easily.



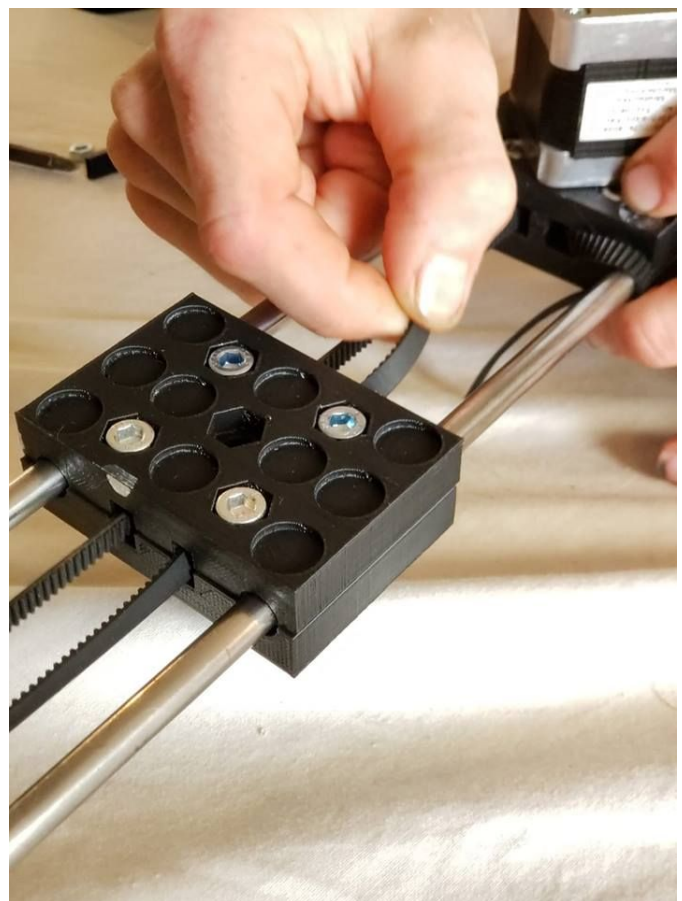
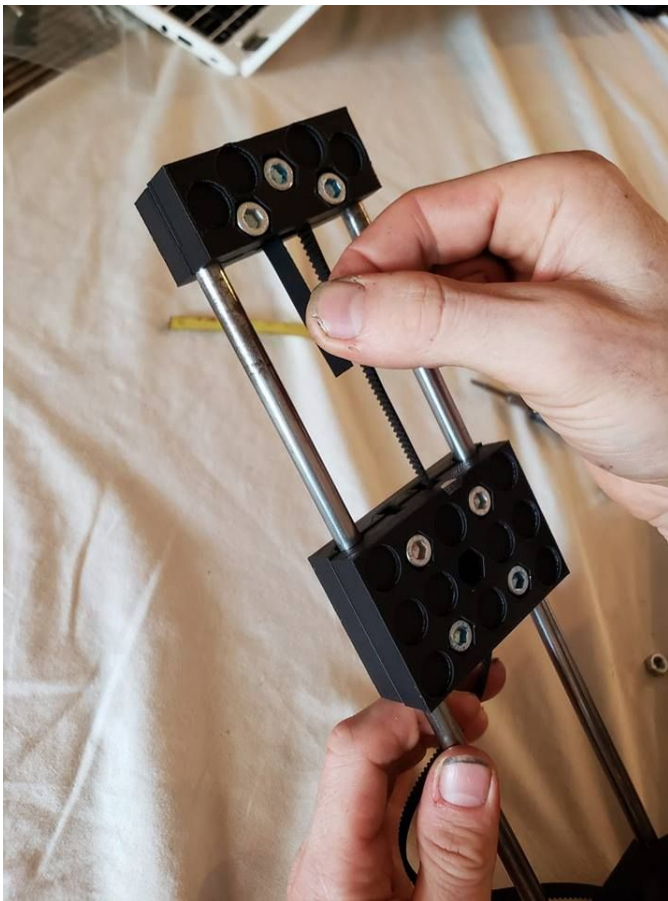
III. Instructional

Belt Threading > Y-Left/Z-Back Axis

Step 7.2: Thread belt around idler bearing (continued)

When successfully around the bearing, pull the belt through.

Thread the belt through the remaining smaller hole of the carriage piece.



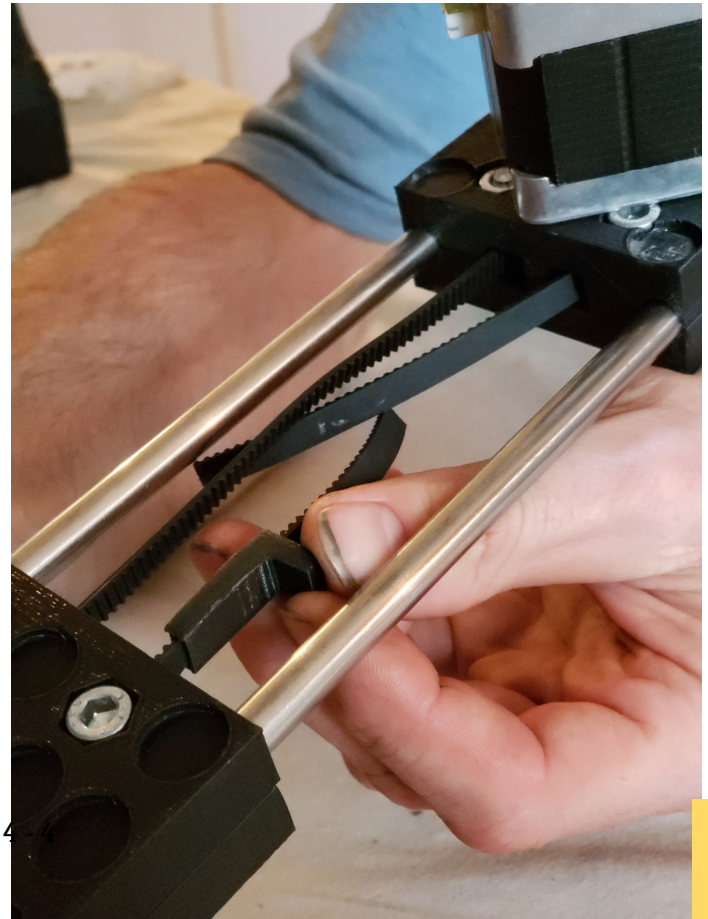
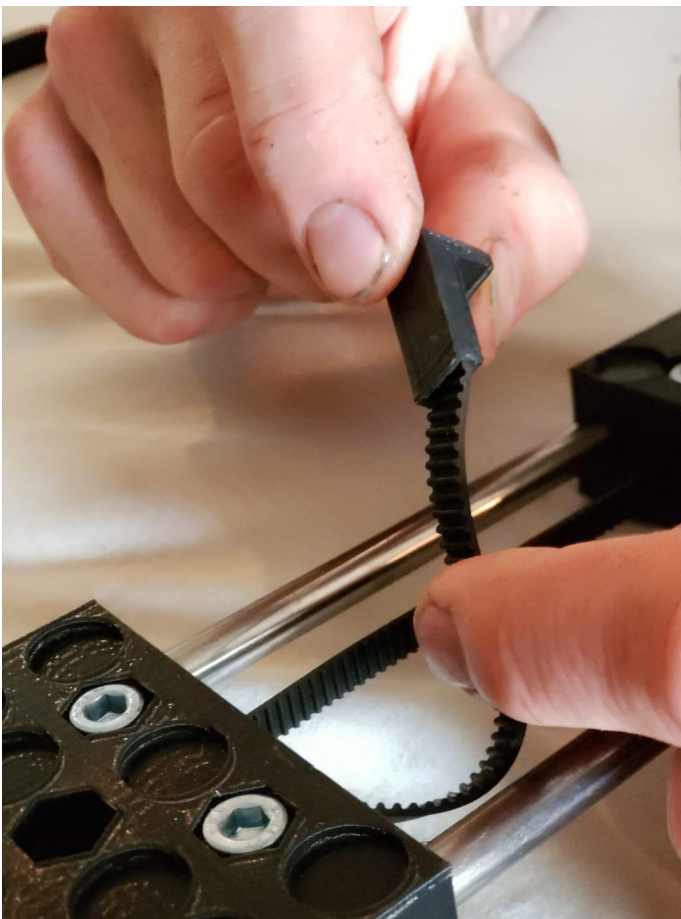
III. Instructional

Belt Threading > Y-Left/Z-Back Axis

Step 7.3: Thread belt through one peg

Thread the belt end through one of the pegs. The “foot” of the peg should be turned away from the toothed side of the belt.

Quick check: *The foot should face away from the center of the axis assembly. The thread should be passing through the carriage before it gets put through the foot.*



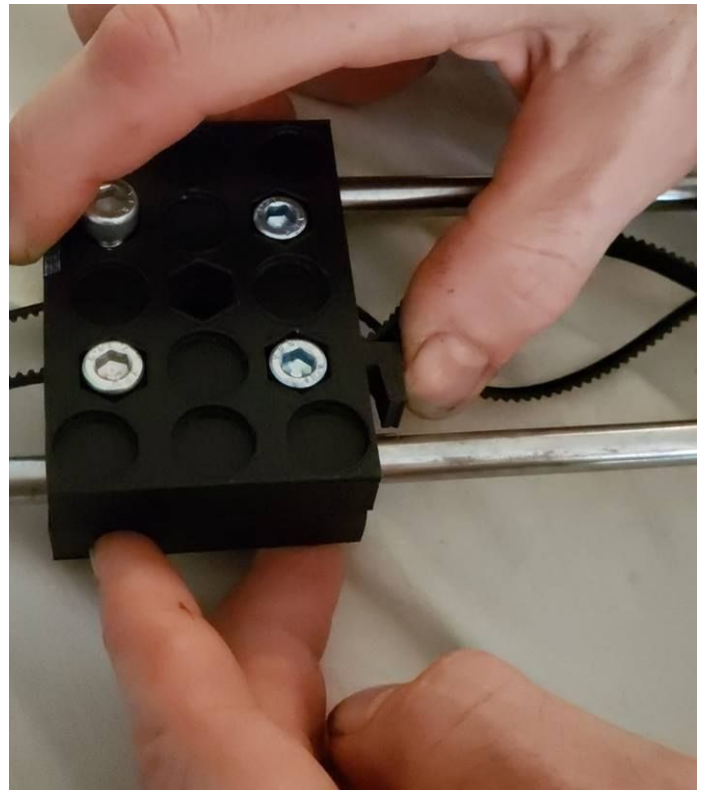
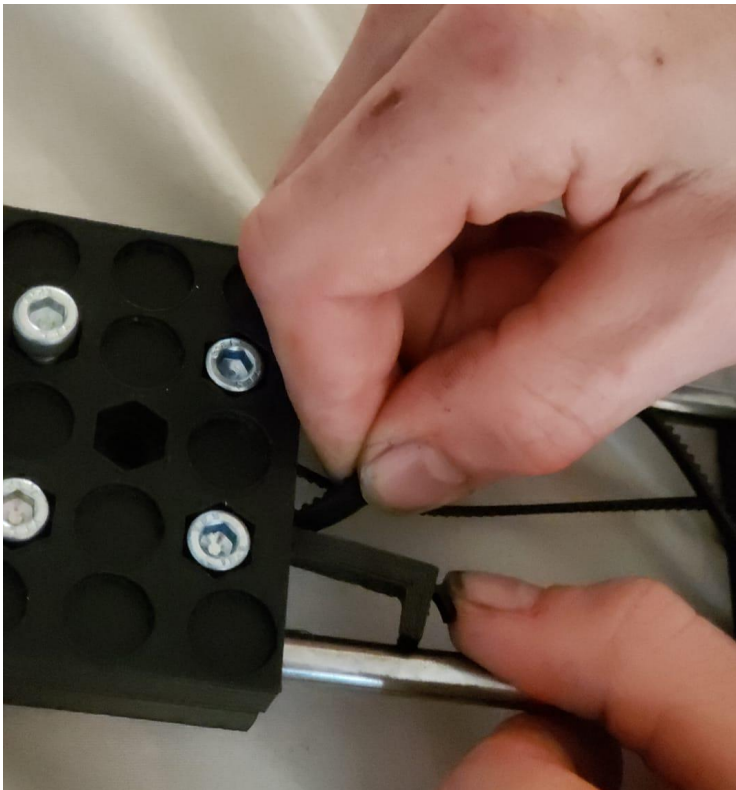
III. Instructional

Belt Threading > Y-Left/Z-Back Axis

Step 7.4: Tighten first peg into carriage

Insert the loose belt end into the belt hole underneath the belt peg, approximately $\frac{3}{4}$ ", then simultaneously insert the peg into the belt hole.

Then tighten down that peg.



III. Instructional

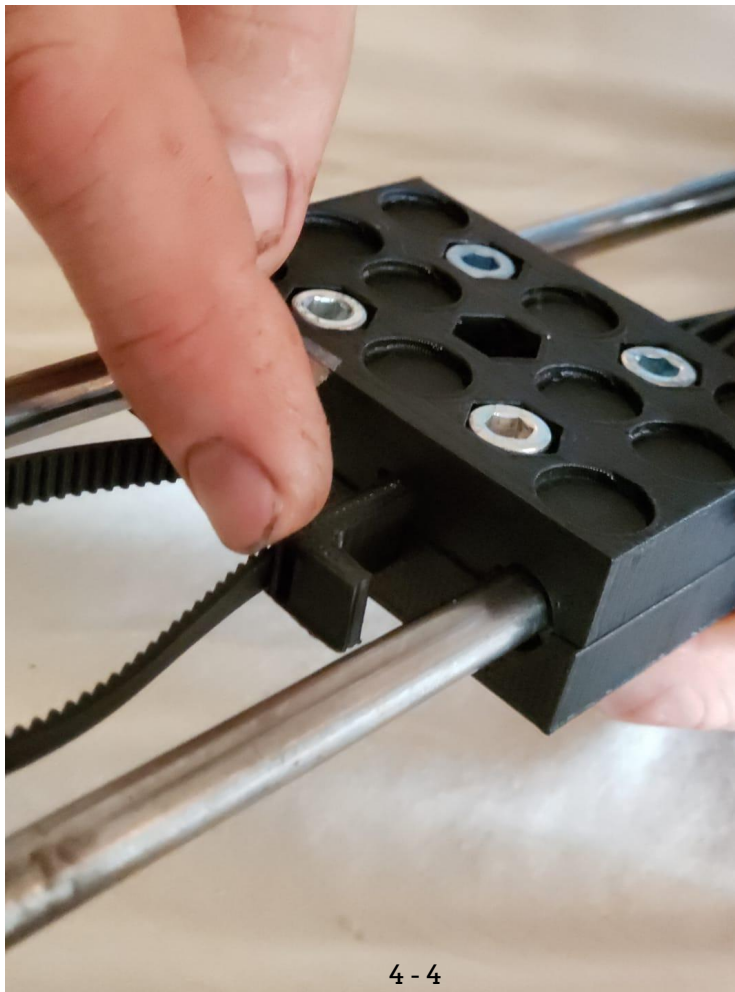
Belt Threading > Y-Left/Z-Back Axis

Step 7.5: Tighten second peg into carriage

Insert a second peg into the other side, directly across from the first peg.

Insert it loosely, then tighten the belt by hand by pulling it farther through the first peg just as much as can be done easily.

Leave this second peg loose, it will be tightened after fitting the Axis Assembly to the frame.



4 - 4

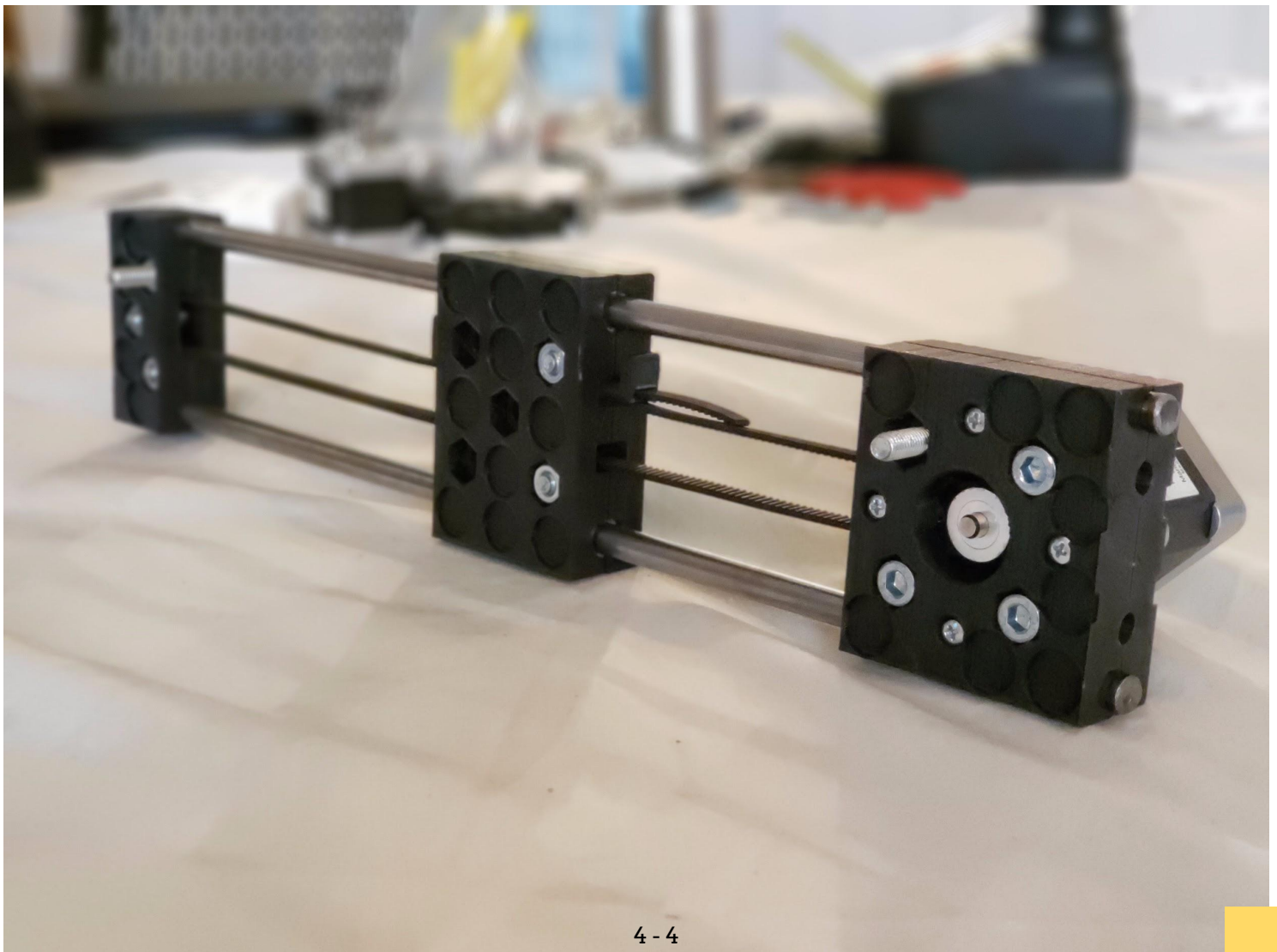
III. Instructional

Y-Left/Z-Back Axis

Step 8: Label Y-left Axis

Your completed Axis Assembly is ready.

Use masking tape and a marker (or other labeling system) to label the Axis Assembly (as the Y-Left or Z-Back Axis) and put it aside for later use.



III. Instructional

Y LEFT ONLY, SKIP FOR Z

Step 9.1: Gather an end stop holder piece and an end stop

Step 9.2 Slide the end stop into the end stop holder piece

Slide into the edge-crevices inside of the end stop holder piece.

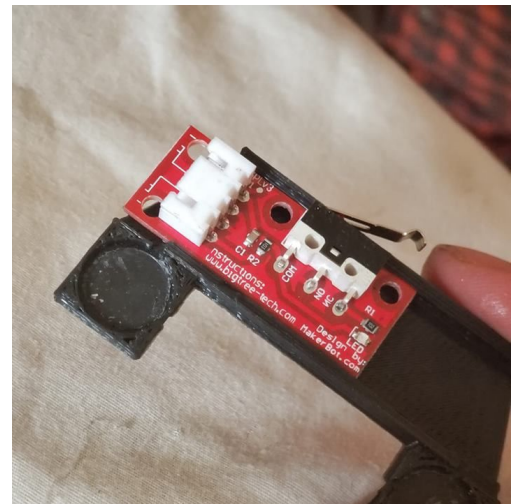
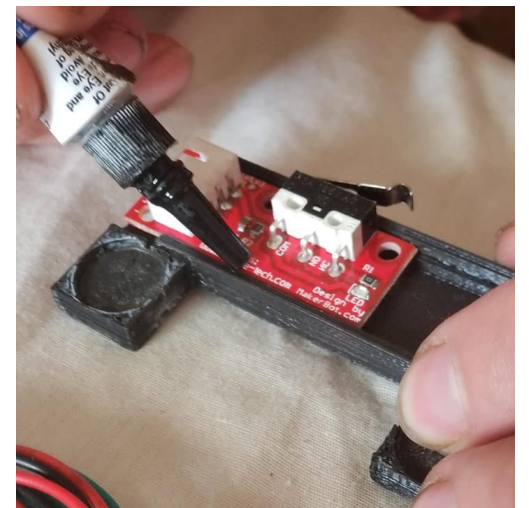
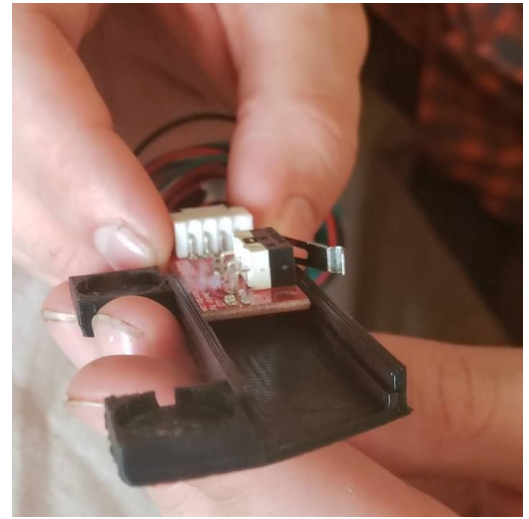
The switch arm should face away from the mounting legs and towards the arrow head that the end piece seems to be making as shown.

The bottom of the end stop should be flush with the bottom of the end stop holder piece.

Step 9.3: Super glue the end stop to the end stop holder piece

Dab super glue along the side edges of the crevice that the board slides through.

Do not put super glue on the metal switch of the end stop.



III. Instructional

Y LEFT ONLY

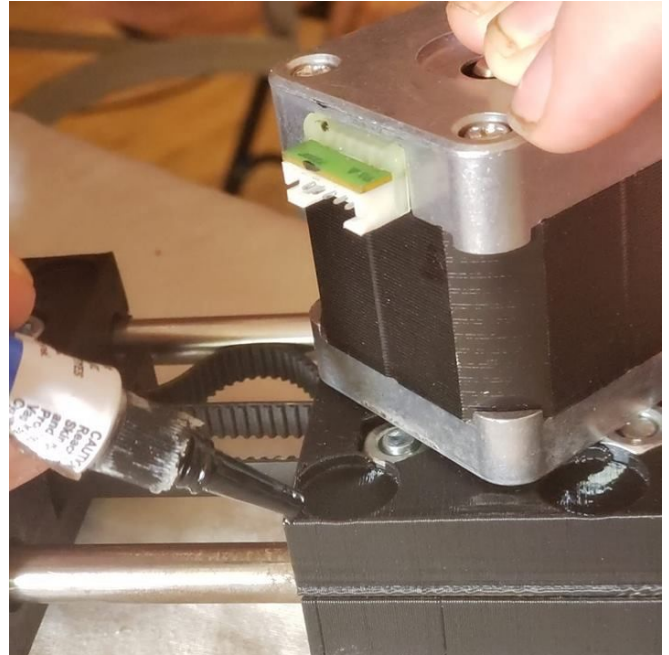
Step 10.1: Prep the Y-axis to have the end stop holder piece super glued to it
Dab super glue into the motor holder piece holes facing towards the belt. These will be the top left and bottom left holes if the motor piece is facing you.

Step 10.2: Super glue the end stop holder piece to the Y-axis motor holder piece

Press and hold the end stop holder piece to the motor holder piece.

Step 10.3: Add super glue along the edges of the end stop holder piece and the motor holder piece.

Adding this super glue here will make the connection a little stronger.



III. Instructional

Y-Left/Z-Back Axis

Step 12.3: Quality Control

- ☐ Belt moves smoothly along axis rods
- ☐ Rods are flush with the idler bottom
- ☐ The idler and carriage bolt heads are on the same side
- ☐ Toothed side of belt is facing inwards

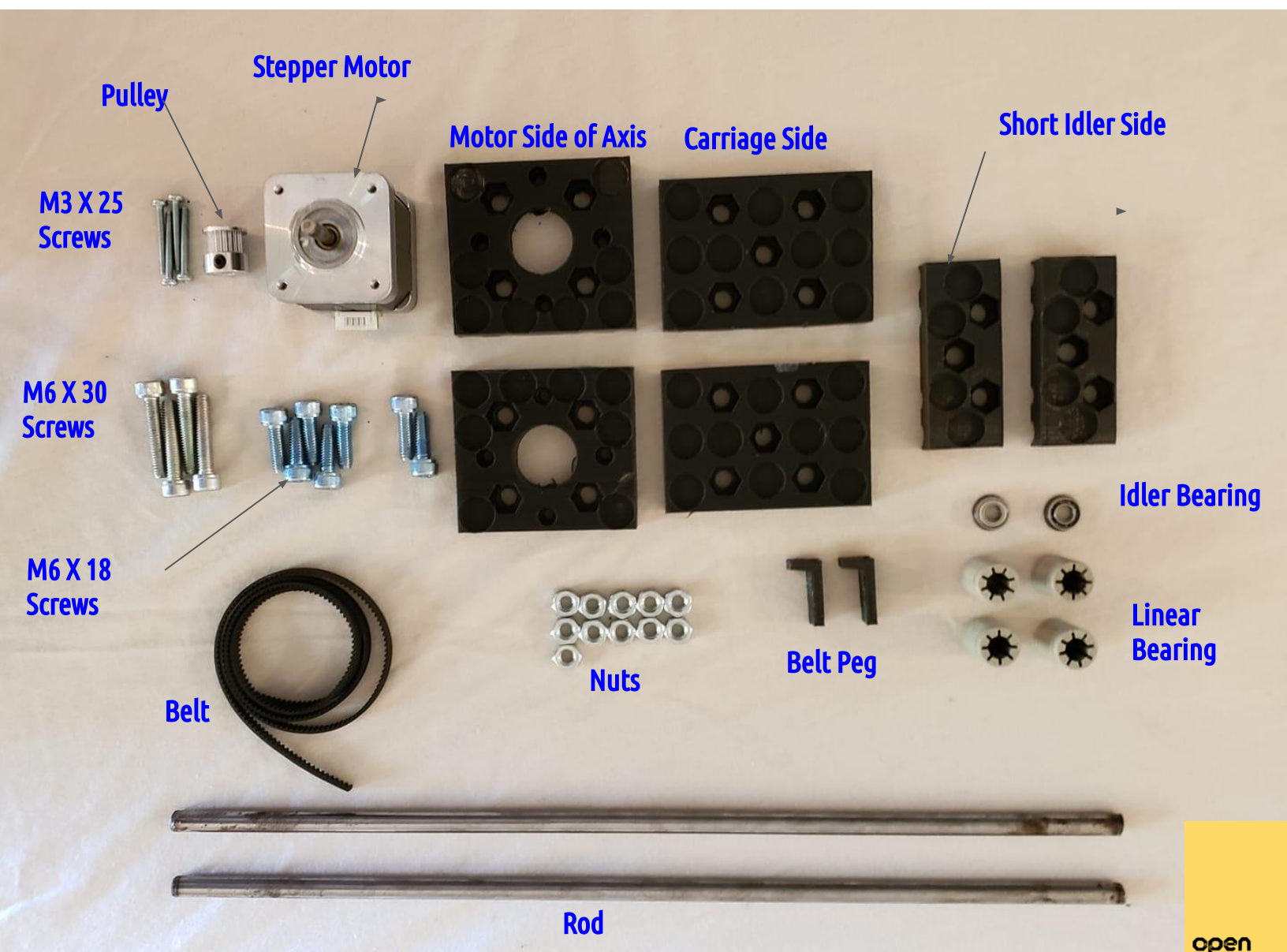
III. Instructional

Y-Right/Z-Front Axis - Parts

These are the parts you need to build a y-right or Z-Front axis.

Note that the instructions for the Y-right and Z-Front axis are meant to be followed twice: once to build the y-right axis, and again to build the Z-Front axis. These parts are identical.

You can choose to make both of these axes at the same time, or you can make them one at a time.



III. Instructional

Y-Right/Z-Front Axis

Tools & Supplies

1 Power Drill

1 Box Cutter or Razor Blade

1 Allen Key -- Metric 5

1 Allen Key -- Metric 2

1 Pair of Needle nose Pliers

1 Piece of Sandpaper grit 220

1 Screwdriver Phillips

1 Spray of oil

Scissors for electrical tape

Drill bit to clean up the plastic prints have imperfections

We recommend 2 people to build an axis. Have an extra power drill and screwdriver for each additional person building.

III. Instructional - Preparation

Y-Left/Z-Back Axis

Step 8.1: Setup & Prepare

Get the correct pieces out.

Step 8.2: Clean up carriage half-pieces, idler half pieces, and motor holder half pieces of big and small holes

Use a box cutter or razor blade to clean out larger curves so that there's no extra plastic covering the holes.

You can also use a power drill with a drill bit to poke through holes and ream the sides of the holes. This means you put the drill bit through a hole, turn on the power drill, and applying pressure to the sides of the hole so that the twisting of the drill whittles away the sides of the hole.

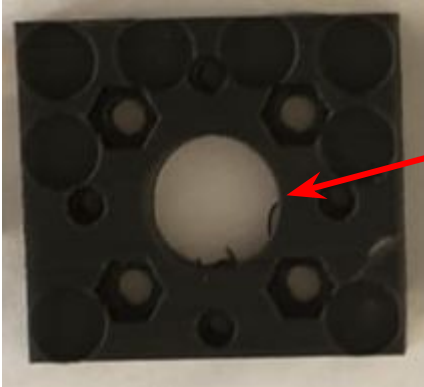
Use a drill bit to thresh out the small holes to get rid of any plastic that covers the hole. Hole should be smooth.

This ensures that rods can be inserted and held in the plastic pieces, and that belts do not rub against the plastic pieces .

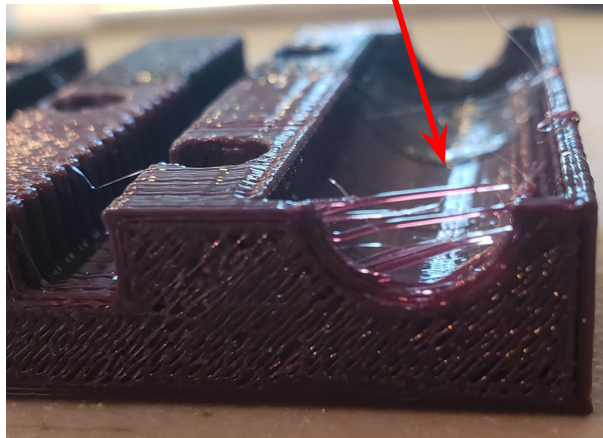
See next page for pictures.

III. Instructional - Preparation

Y-Left/Z-Back Axis



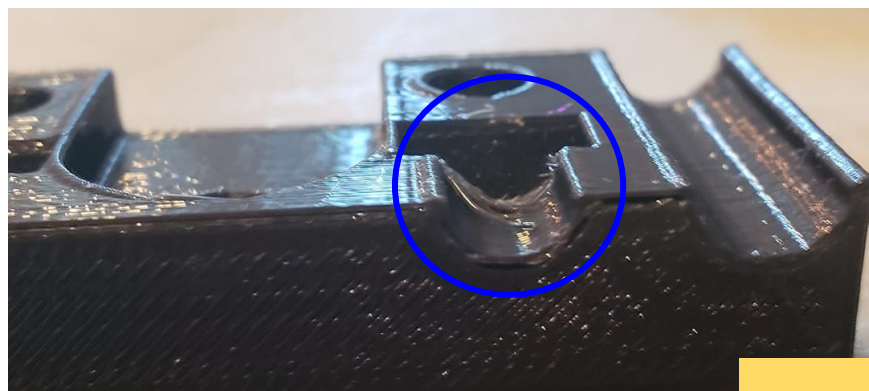
Clean this up



Carriage half piece with excess plastic
threading



Motor Holder Half Piece with excess
plastic



Motor Holder Half Piece cleaned

III. Instructional - Preparation

Y-Right/Z-Front Axis

Step 1.3: Plastic Parts Quality Control

- ❑ Each Carriage Half Piece has 4 clean semi-circles
- ❑ Each motor holder piece has 6 clean semi-circles
- ❑ Each idler piece 8 clean semi-circles

III. Instructional

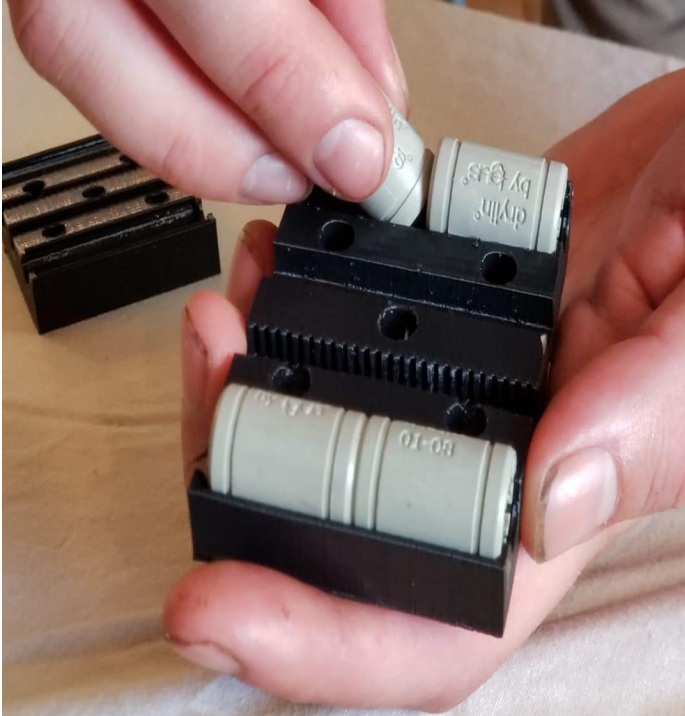
Carriage > Y-Right/Z-Front Axis

Step 8.4: Test bearings fit

Place the four bearings in one side of the Axis carriage. Add another side of carriage on top and hold them together.

It needs to be tight enough for a good print, but loose enough for the carriage to slide on the rods.

Follow the next step to test that the bearings fit well.



III. Instructional

Carriage > Y-Right/Z-Front Axis

Test Method 1: Carriage Shake

This test ensures that the bearing is tight enough.

Put bearings in carriage. Close the carriage with your hand and compress the carriage piece tightly. Shake it. If the bearings shake inside the carriage, you will need to adjust the bearing size so that it is a snug fit inside the carriage and doesn't shake.



4 - 4

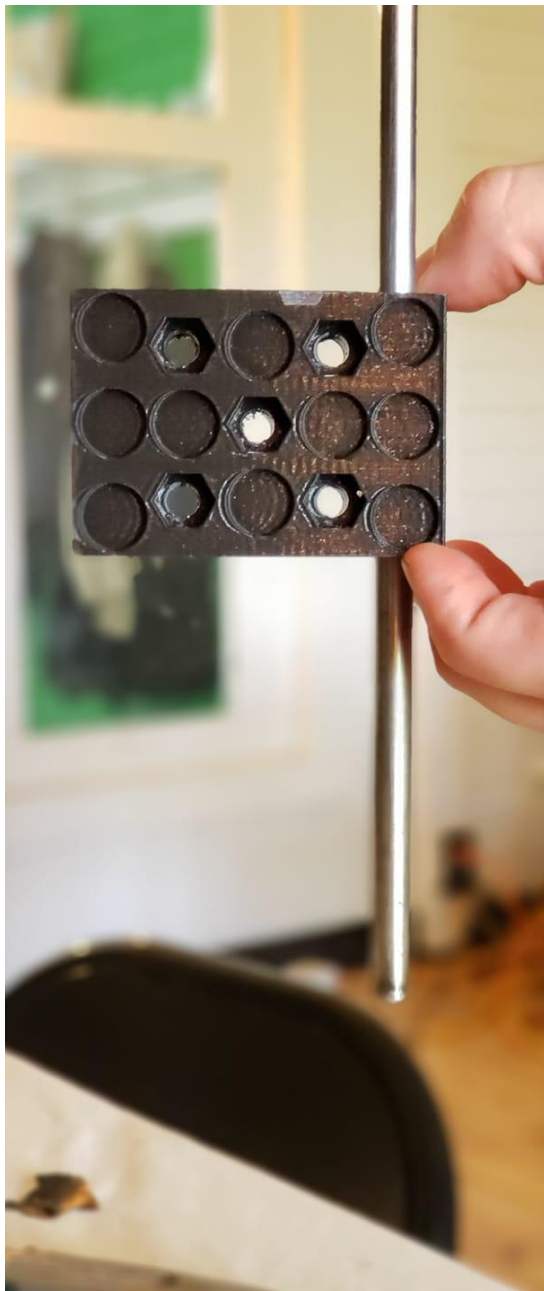
III. Instructional

Carriage > Y-Right/Z-Front Axis

Test Method 2: Rod Slide

This test ensures that the bearing is not too tight.

Put the rod through the carriage. Turn the carriage so that the rod is vertical. Rod should slide or drop through as you hold the carriage.



Motors will be pulling a belt attached to this carriage with 20 pounds of force. So the rod shouldn't just slip through the carriage super fast; instead, **there should be a little resistance**. It's good if you have to shake the carriage for the rod to slide through.

If the rod slides too quickly through the carriage without any resistance, you should probably adjust the bearings by making them tighter.

It's likely the rod won't be adjusted exactly right. See next step to use tape to adjust the tightness.

III. Instructional

Carriage > Y-Right/Z-Front Axis

Step 8.5: Adjust bearings if needed

Wrap each bearing in one layer of electrical tape. Tape so that it just barely overlaps with itself. We recommend electrical tape because it's on hand in this build, but any tape will work.

Repeat as necessary.

Why does tape make it tighter?

If the rod slips through really quickly, then adding tape to the bearings will give it more resistance, even though the rod is always slipping through the same place.

Why can tape make it too tight?

Too much tape will compress the bearing and keep it from sliding.



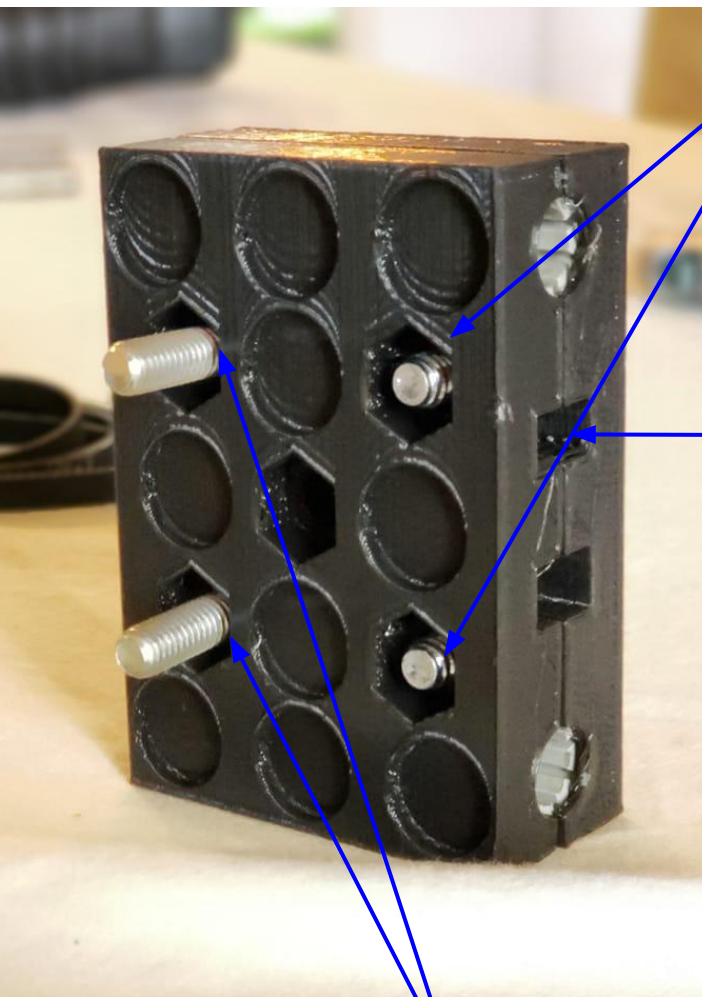
4 - 4

III. Instructional

Carriage > Y-Right/Z-Front Axis

Step 2.1: Assemble Carriage

Four bolts will be used to secure the carriage piece as shown. Note the proper alignment of the rectangular belt holes.



M6 X 18 bolts

Note there are 2 different belt holes. One is small and ribbed. The other is larger and smooth. The belt peg goes into the small, ribbed hole.

Correct alignment of holes

M6 X 30 bolts

Incorrect alignment of holes mismatches the small and large holes



III. Instructional

Carriage > Y-Right/Z-Front Axis

Step 9.2: Tighten bolts with nuts

Use an electric drill on the lowest setting to secure nuts to the four bolts.



You've completed an axis carriage.

Optional: Repeat the Carriage Assembly three more times so you have a total of four built for all the Axis Assemblies.

III. Instructional

Idler > Y-Right/Z-Front Axis

Step 10.1: Insert M6x18 Bolt

Begin assembling the idler by placing a M6x18 bolt through one side.



Step 10.2: Insert two bearing pieces

Insert the bearings one on top of the other on the bolt. Flanges should be facing outside, as shown.



Step 10.3: Identify how to put the idler piece together correctly

Make sure the idler pieces align correctly. These circular holes for the rods should line up with each other.



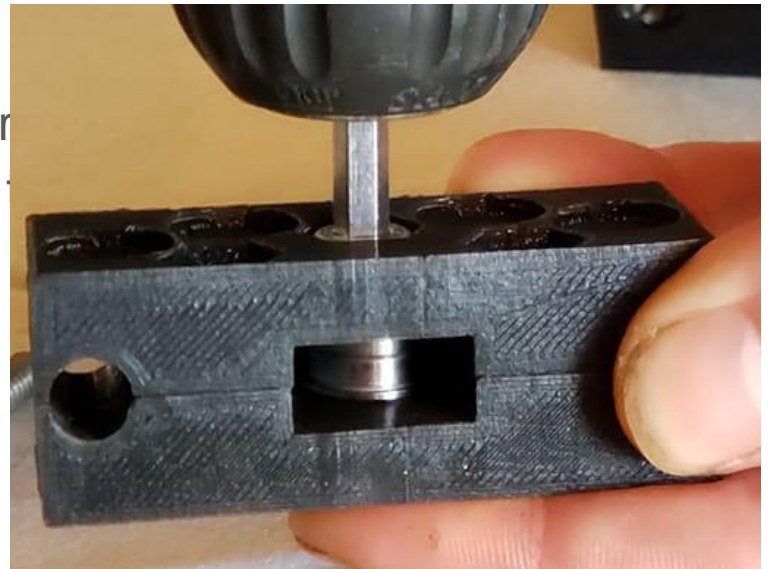
III. Instructional

Y-Right/Z-Front Axis

Step 10.4: Secure bearing with one M6x18 bolt

Put a nut on the back of the bolt and secure fully with an electrical drill on the lowest setting.

It doesn't matter which way you screw bolts into the idler. Just make sure all bolts are all facing the same way.



Step 10.5: Put 1 m6x30 screw into correct hole

Orient the idler piece so that the large rectangular hole is facing to your left. Insert one M6x30 screw into the top hole so that the screw end points facing you. Secure loosely with a nut on the opposite side -- because you will be attaching this to the frame later.



III. Instructional

Y-Right/Z-Front Axis

Step 10.6: Secure a M6x18 bolt through the last hole using a nut and the electric drill.

Screw the nut on, not so that it's tightened, but kept very loose as shown. We will loosen this piece later on to attach to the frame.



Your idler is ready to go. Put to the side and continue on to assembling the motor.

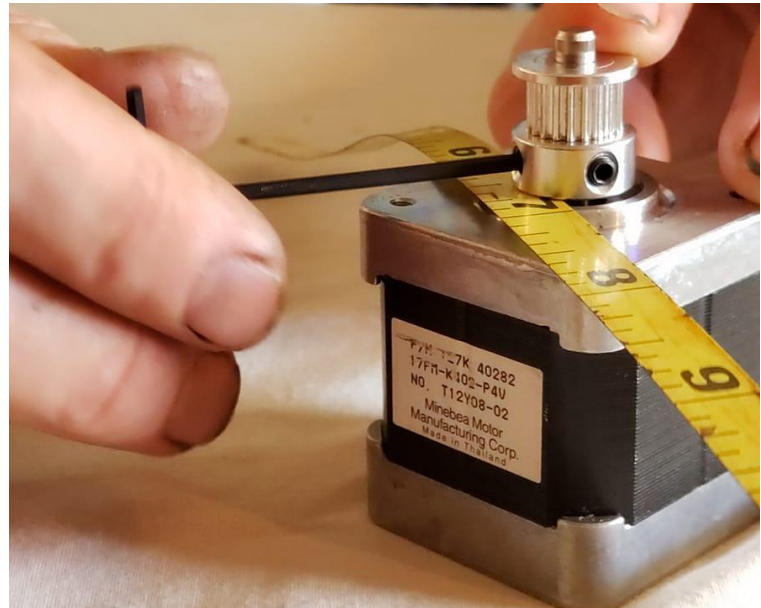
III. Instructional

Motor > Y-Right/Z-Front Axis

Step 11.1: Put pulley on motor

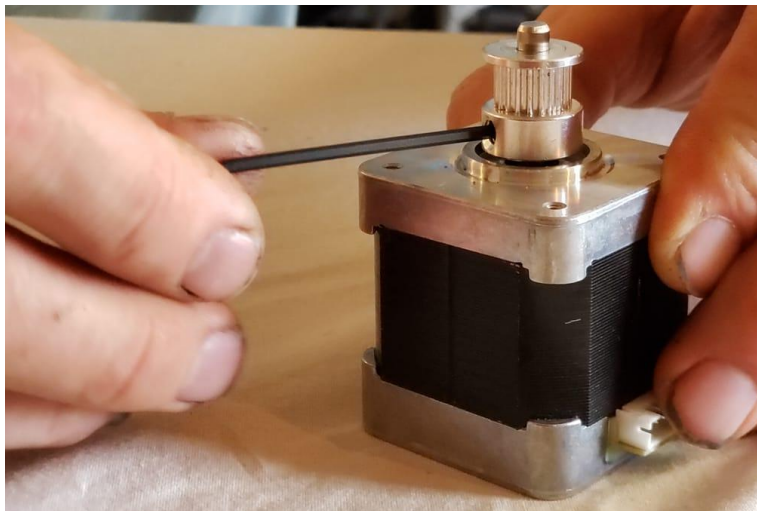
Place the pulley on the motor using a thin spacer (provided) that the pulley rests upon. Tighten the first set screw with an Allen wrench.

The spacer prevents the pulley from rubbing on the motor housing, and assists in aligning the pulley gear to the belt channel.



Step 11.2: Tighten pulley set screw onto motor shaft

With the first set screw tightened, you can remove the spacer. Then tighten the second set screw.



III. Instructional

Motor > Y-Right/Z-Front Axis

Step 11.3: **Quality Control Check**

Flip the Allen wrench and tighten down the set screw as far as you can, pressing down on the wrench near its midpoint for greatest pressure. When you hear the set screw "pop.", you can be sure you've set it as tightly as possible.

The pop sound occurs when the small screw head slips along the motor shaft. It doesn't get any tighter than that. This tight fit ensures the belt will be pulled effectively by the motor shaft.

It's also OK if you don't hear a pop sound. As long as you've tightened the screw as hard as you can, then it should be fine.

In the next steps you will be assembling the motor pieces.

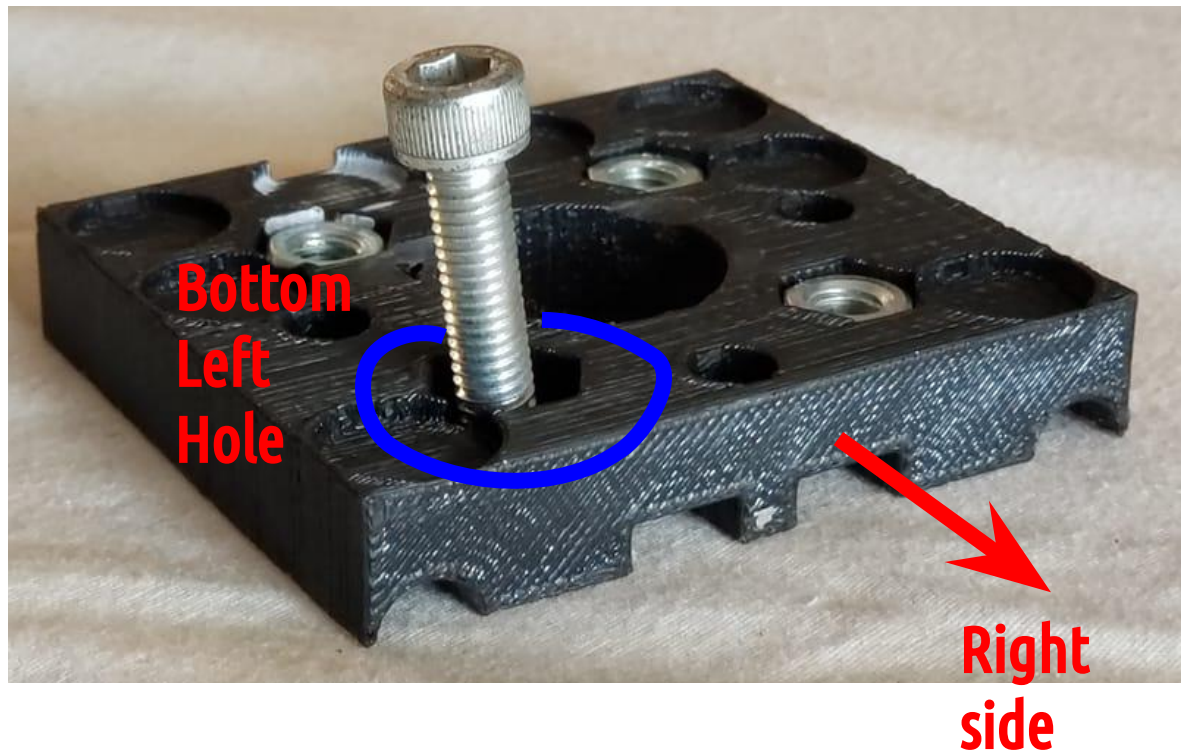


III. Instructional

Y-Right/Z-Front Axis

Step 10.1: Setup Motor half-piece with three nuts, one M6x30 bolt

Place bolt so that it's popping up as shown. Put nuts in on other 3 holes. Bolt goes in the bottom left hole, where the right side of the motor half-piece has the belt drive. Orientation matters here.



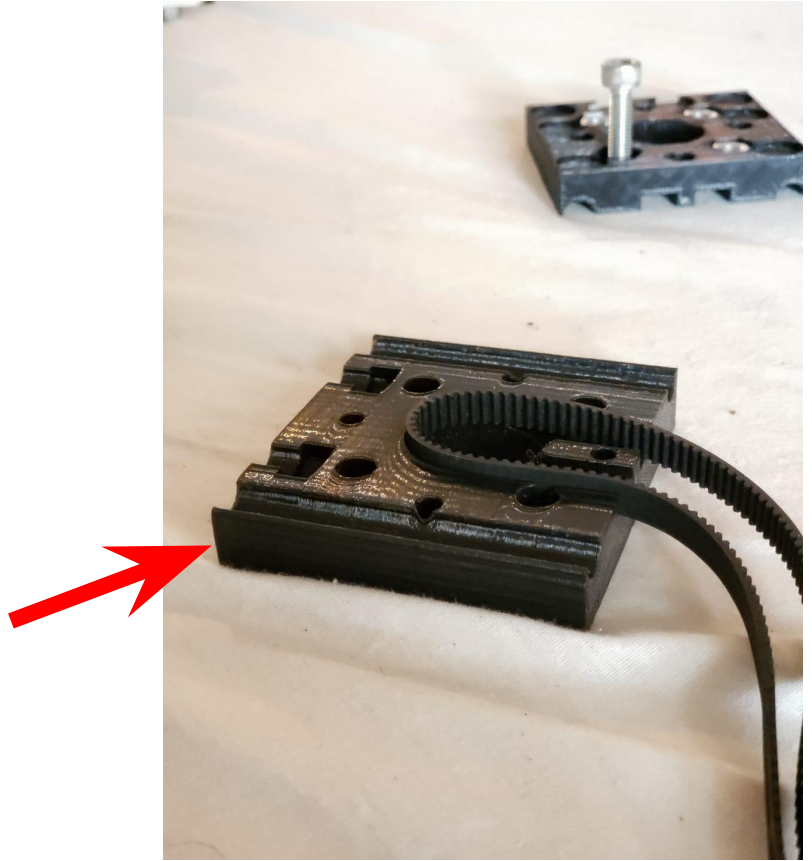
When you put the motor on top of the motor half-piece, this lets you access the screws because the motor will be covering the pieces later on. This will make sense in pictures down the line.

III. Instructional

Motor > Y-Right/Z-Front Axis

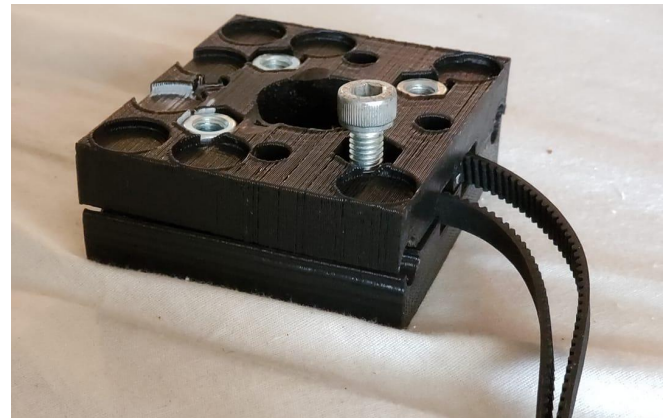
Step 10.2: Set up the other half piece with the belt set inside

There's a nice channel for the belt on the other half piece. Insert the belt with the toothed side facing inwards.



Step 10.3: Close up the two half pieces

Be gentle with the closed half piece because you want the belt to remain in place without slipping.



III. Instructional

Y-Right/Z-Front Axis

Step 10.3: Fasten small bolts into motor through motor piece

Hold the Motor+Motor Piece with the motor resting in your hand.

Put 4 M3x25 screws into the smallest holes and tighten with a phillips screwdriver.

Double check that the bolts are still in the carriage.

Do not fasten bolts too tight.

Nuts should be in nut catchers on the motor side.

Place the motor so that, when the motor is facing you, the cable plug of the motor is facing towards the belt drive hole.

Additionally, the motor plug should be facing in the opposite direction of where the long m6x30 bolt is.

This makes it so the wires plugged into motor can be wrapped nicely.

Step 10.4: Fasten 3 m6x18 bolts into motor piece

Motor shaft should be visible at the same time as seeing



III. Instructional

Y-Right/Z-Front Axis

Step 10.5 Motor Piece Quality Control

Have at least 3 screws on the motor, and 4 screws on the motor piece, plus the bolt that's facing out.

Step 11.1 : Assemble Y-Right Idler piece

Place the rods in their respective holes in the idler piece, ensuring that they are flush with the outside. The M6-30mm bolt should be on top, with the arrow of bolts facing away from the rods.

If it is difficult to fit the rods on, just loosen the bolts until the rods slide in. Then re-tighten.



III. Instructional

Y-Right/Z-Front Axis

Step 11.2: Put carriage assembly onto rods

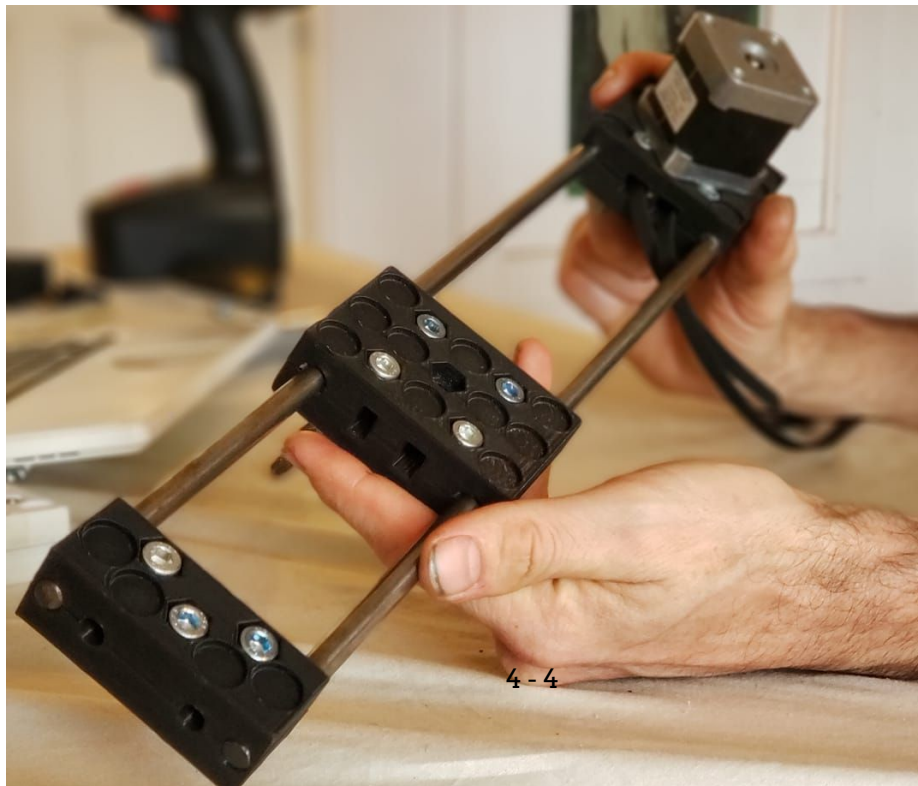
Put carriage onto rods such that the long bolts should be closer to the idler piece when the carriage is put in.

This makes it so that the carriage can travel farther and therefore increases the printing space.

Step 11.3: Put motor piece onto rods

Push motor piece onto rods such that motor is facing the heads of the bolts of the carriage you just put in [see picture].

If it is difficult to fit the rods on, just loosen the bolts until the rods slide in. Then re-tighten.



4 - 4

III. Instructional

Y-Right/Z-Front Axis

Step 11.4: Quality Control Rods are Flush

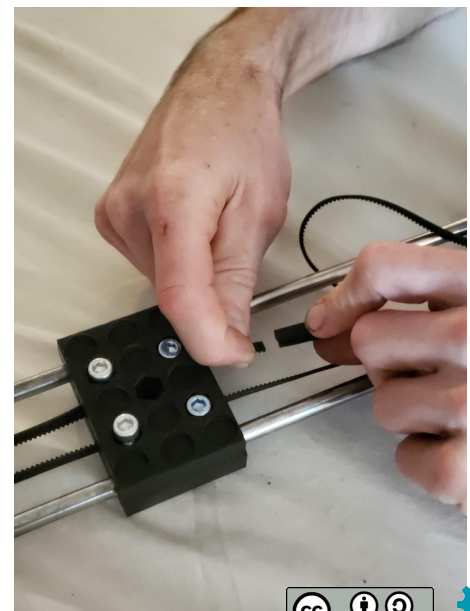
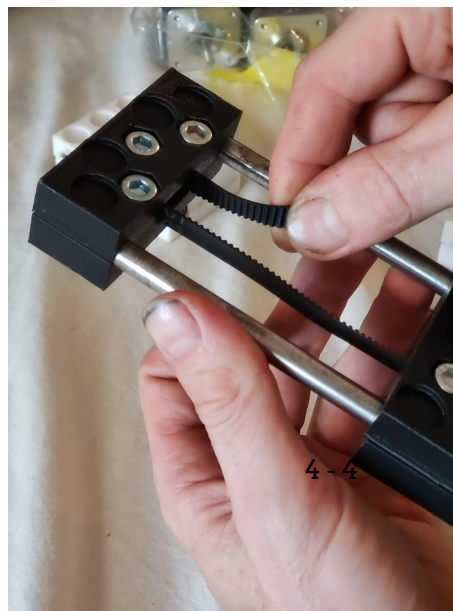
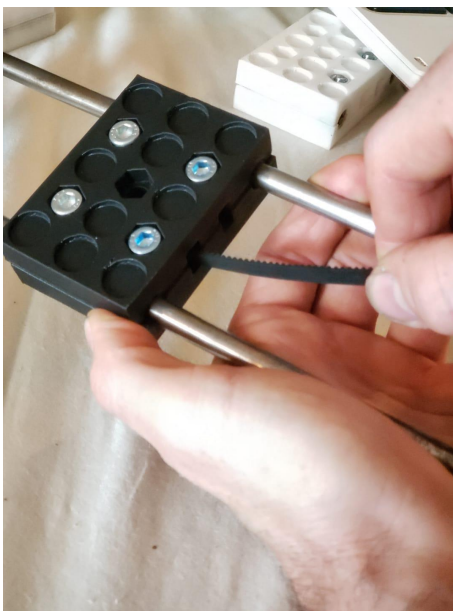
Check the idler that the rod is still flush. (see image on previous page)

Step 11.5: Thread belt into carriage

You are going to tighten the belt through the motor piece, carriage, and idler. In this step, you will end with the two loose belt ends meeting at the small space of the carriage. This will put you in the right situation to tighten the belt properly on the carriage.

Look at where the large and small space for the carriage is. Aim to put the belt through the carriage such that the two loose ends meet at the small space in the carriage. This means you put the belt through the ... first.

Thread through idler. On the belt catcher side. This is the smaller of the two holes, in which there are teeth on one side. Belt should be looking basically complete.



III. Instructional

Y-Right/Z-Front Axis

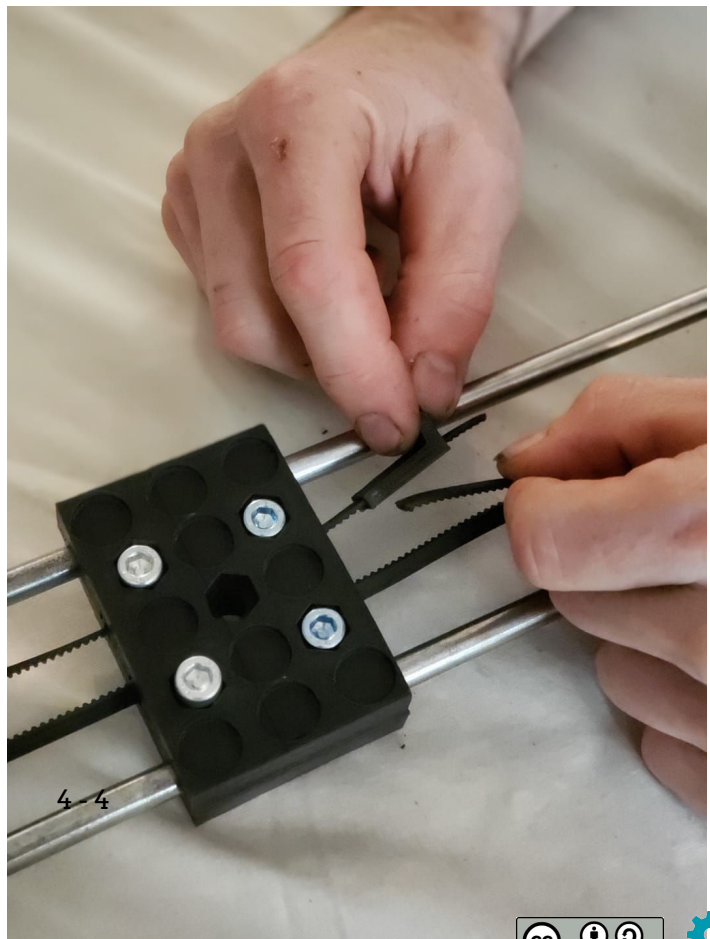
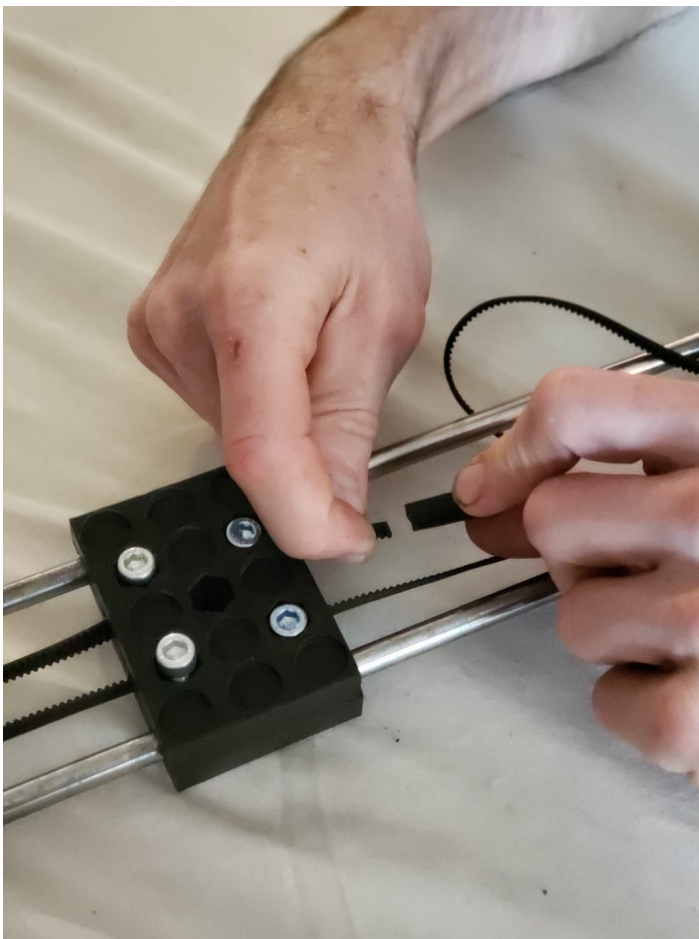
Step 12.1: Tighten Belt into carriage

Thread the end of the belt which you just put through the idler through the belt catcher hole, and pull the end out of the other side.

Thread through the inside of the belt peg with the belt teeth opposite of the foot. The foot will face away from the center of the axis assembly. The belt should be passing through the carriage before it gets put through the foot.

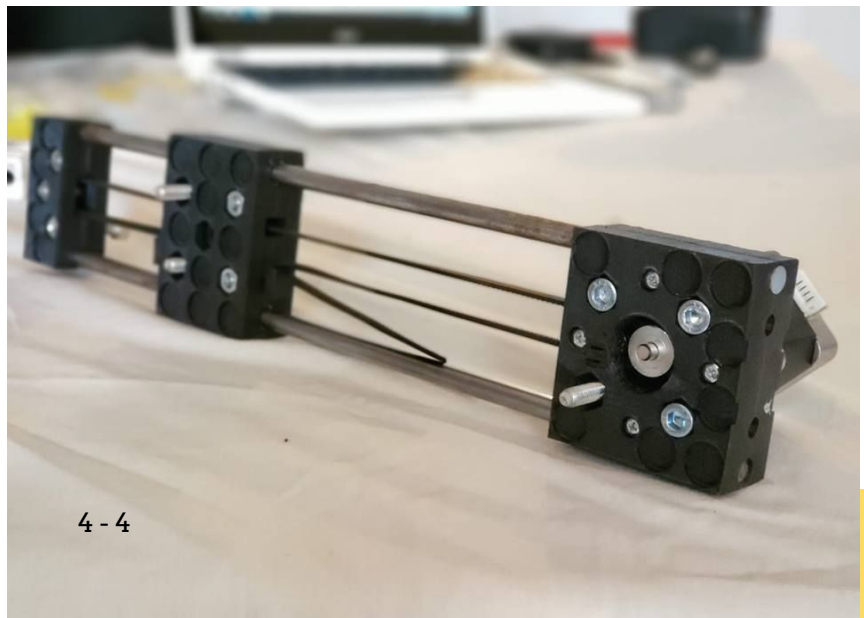
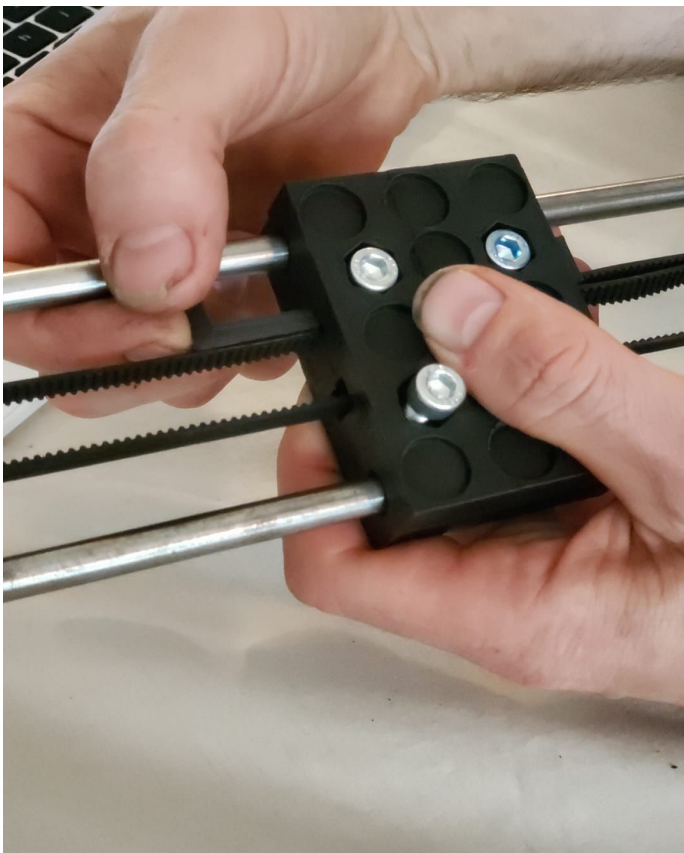
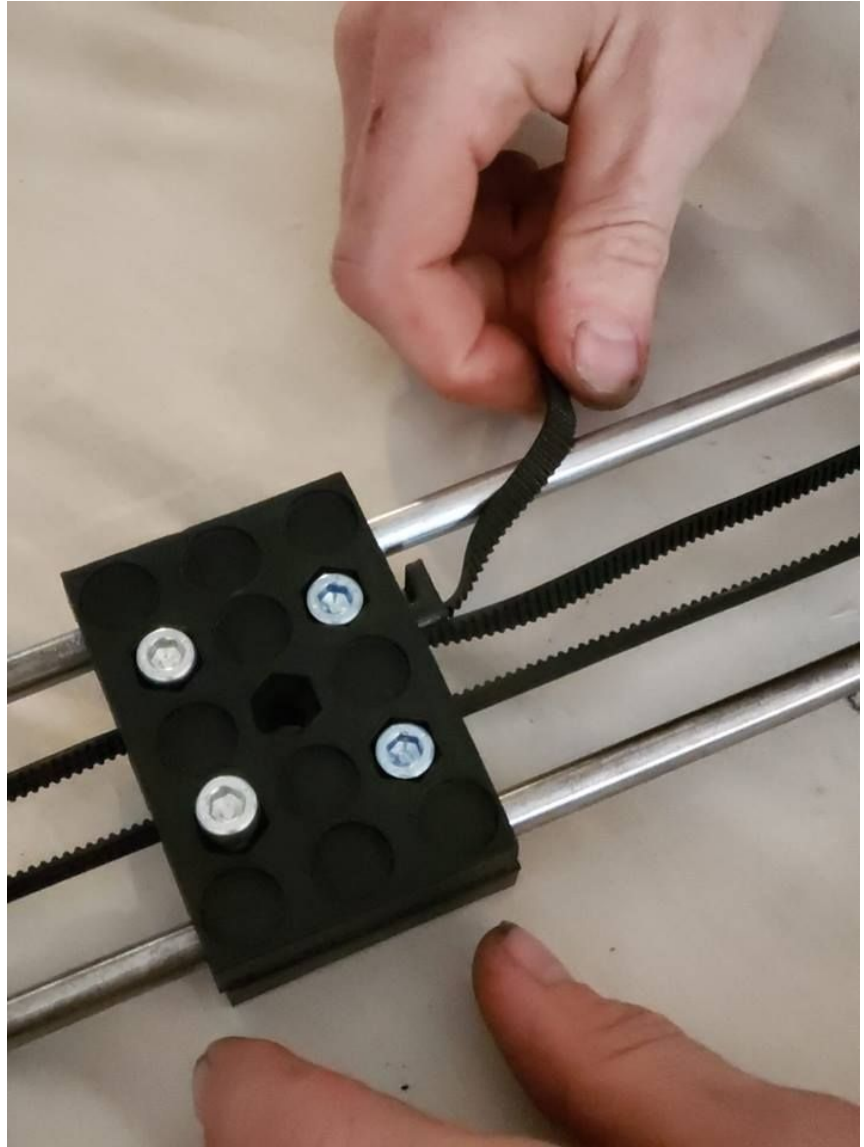
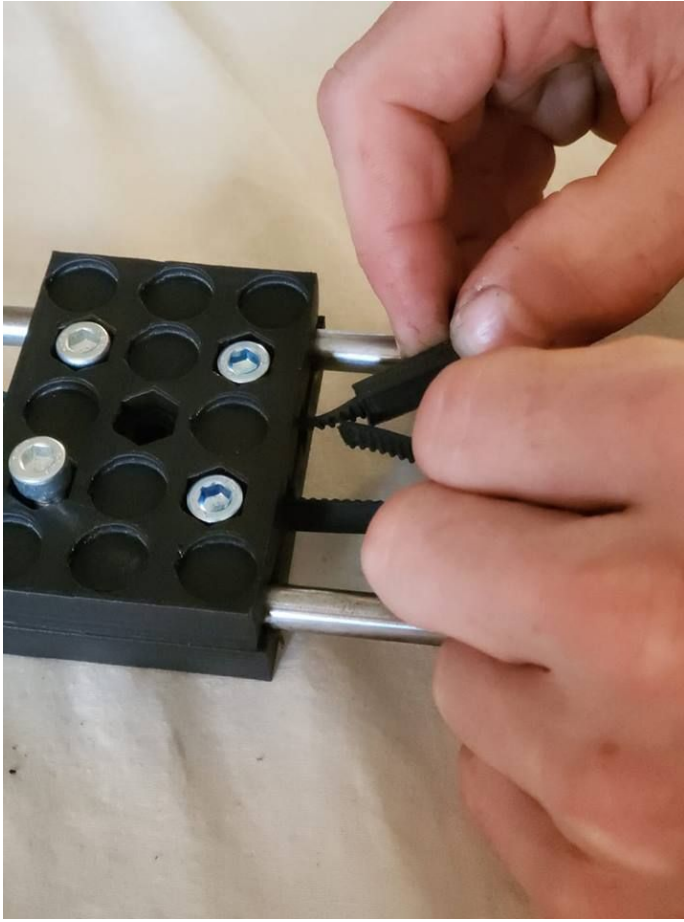
Catch the belt under the peg, approximately $\frac{3}{4}$ " and insert the belt peg into the belt catcher hole.

Place the other peg loosely in the other side for later tensioning.



III. Instructional

Y-Right/Z-Front Axis



4 - 4

III. Instructional

Y-Right/Z-Front Axis

Step 12.2: Label the completed axis

Use masking tape and marker to label the axis “Y-Right” or “Z-Front” and put it aside for later use.

III. Instructional

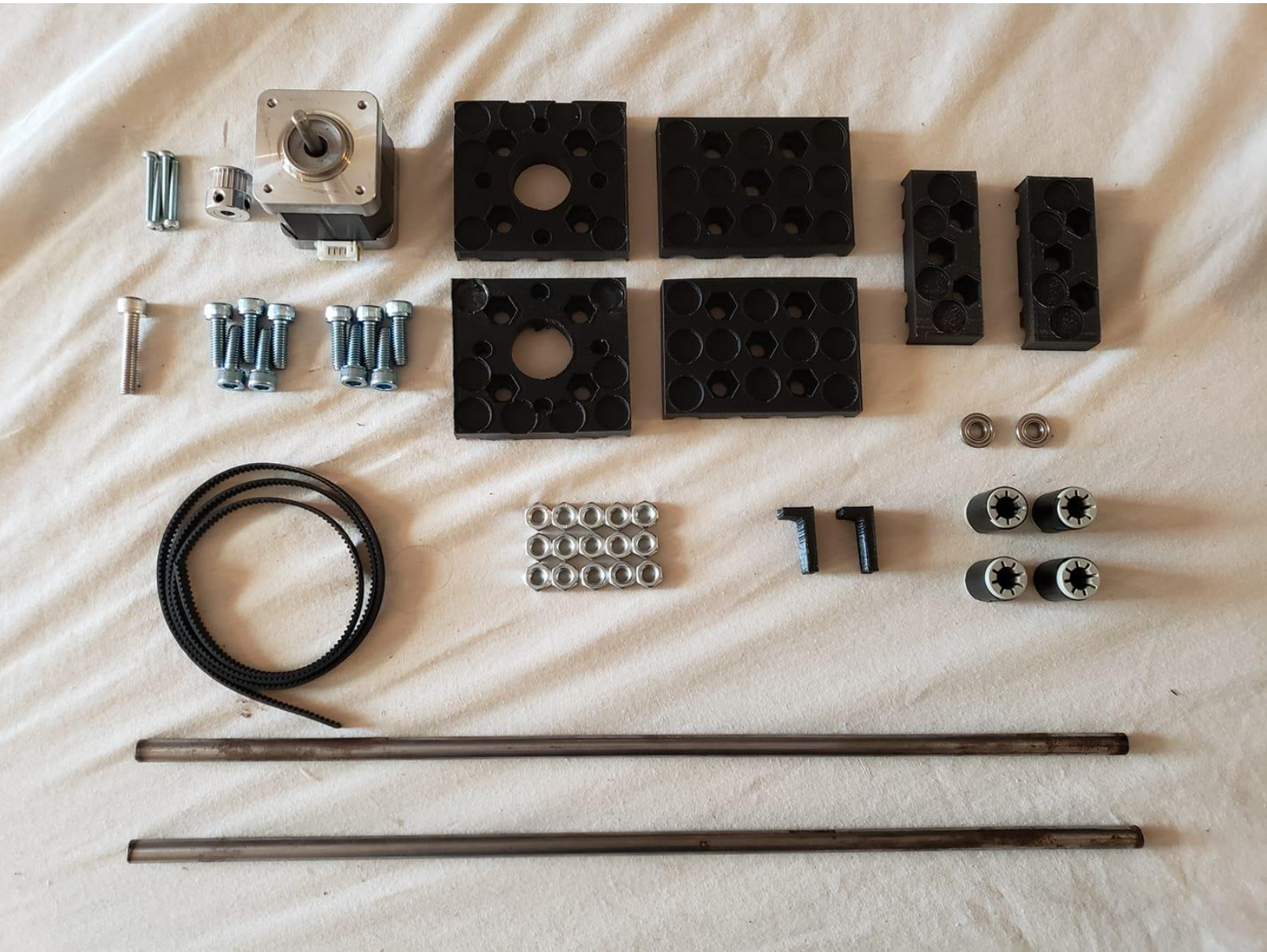
Y-Right/Z-Front Axis

Step 12.3: Quality Control

- ☐ Belt moves smoothly along axis rods
- ☐ Rods are flush with the idler bottom
- ☐ The idler and carriage bolt heads are on the same side
- ☐ Toothed side of belt is facing inwards

III. Instructional

X Axis - Parts



III. Instructional - Preparation

X Axis

Step 13.1: Setup X Axis

Get the correct pieces out.

Step 13.2: Inspect Carriage and other small holes

Use a drill bit to thresh out the small holes to get rid of any plastic that covers the hole. Hole should be smooth.

This ensures that rods can be inserted and held in the plastic pieces, and that belts do not rub against the plastic pieces .

Step 13.3: Inspect Carriages and other big holes

Use a box cutter or razor blade to clean out larger curves so that there's no extra plastic covering the holes.

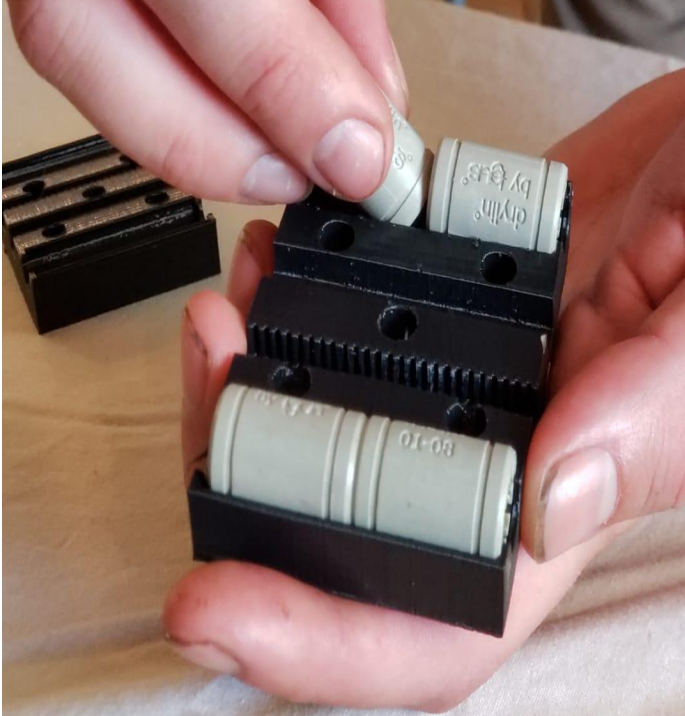
III. Instructional

X Axis

Step 13.4: Test bearings fit

Place the four bearings in one side of the Axis carriage. Add another side of carriage on top and hold them together.

It needs to be tight enough for a good print, but loose enough for the carriage to slide.



III. Instructional

X Axis

Test Method 1: Carriage Shake

This test ensures that the bearing is tight enough.

Put bearings in carriage. Close the carriage with your hand and compress the carriage piece tightly. Shake it. If the bearings shake inside the carriage, you will need to adjust the bearing size so that it is a snug fit inside the carriage and doesn't shake.



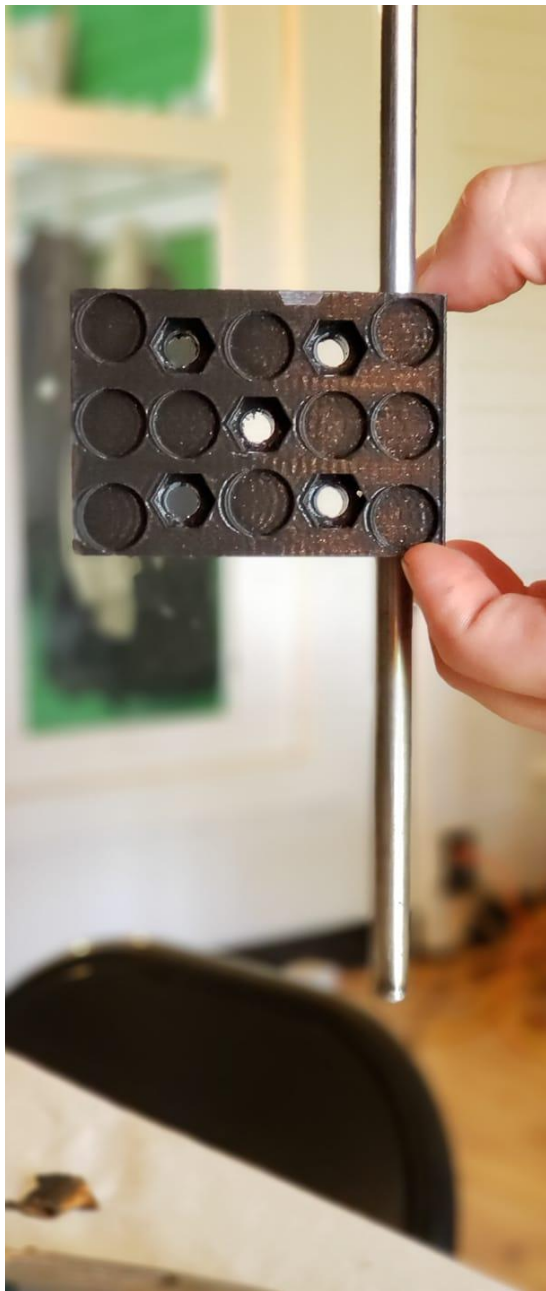
III. Instructional

X Axis

Test Method 2: Rod Slide

This test ensures that the bearing is not too tight.

Put the rod through the carriage. Turn the carriage so that the rod is vertical. Rod should slide or drop through as you hold the carriage.



Motors will be pulling a belt attached to this carriage with 20 pounds of force. So the rod shouldn't just slip through the carriage super fast; instead, there should be a little resistance. It's even OK if you have to shake the carriage for the rod to slide through. It's also OK.

It's likely the rod won't be adjusted exactly right. See next step to use tape to adjust the tightness.

III. Instructional

X Axis

Step 13.5: Adjust bearings if needed

Wrap each bearing in one wrapping of electrical tape. Tape so that it just barely overlaps with itself. We recommend electrical tape because it's on hand in this build, but any tape will work.

Repeat as necessary.

Why does tape make it tighter?

If the rod slips through really quickly, then adding tape to the bearings will give it more resistance, even though the rod is always slipping through the same place. There is more resistance because

Why can tape make it too tight?

Too much tape will compress the bearing and keep it from sliding.

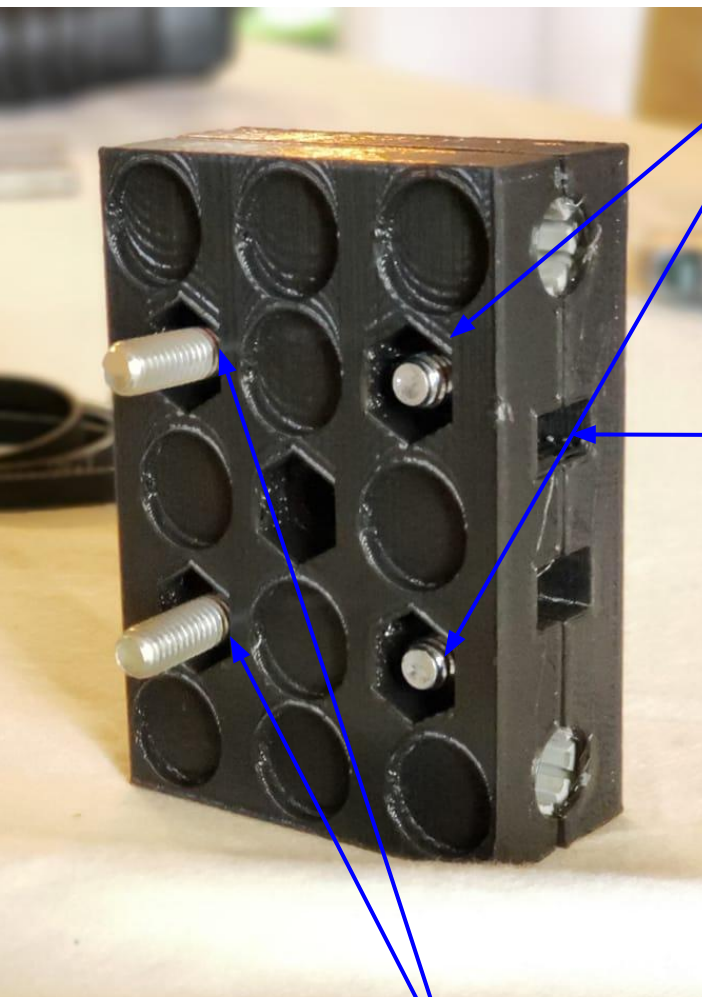


III. Instructional

X Axis Carriage

Step 14.1: Assemble Carriage

The middle holes should fit together as shown in picture.



M6 X 18 bolts

Note there are 2 different belt holes. One is small and ribbed. The other is larger and smooth. The belt peg goes into the small, ribbed hole.

Correct alignment of holes

M6 X 30 bolts

Incorrect alignment of holes mismatches the small and large holes



III. Instructional

X Axis Carriage

Step 14.2: Tighten carriage bolts

Using one nut for each bolt, tighten the bolts with an electric drill on the lowest setting.



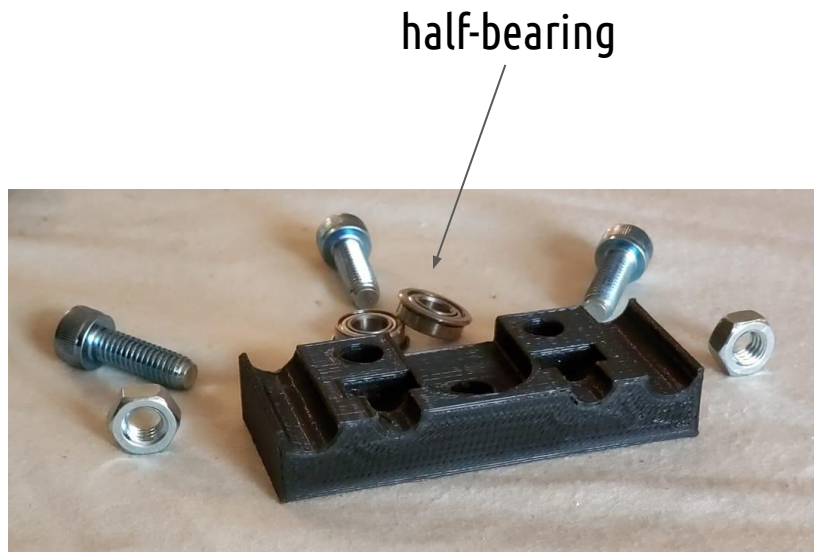
4 - 4

III. Instructional

X Axis Idler

Step 15.1: Assemble the idler

Get [5] M6x18 screws, 2 half-bearings, and 2 nuts to assemble the idler.



Step 15.2: Insert M6x18 bolt

Begin assembling the idler piece by placing a bolt through one side



Step 15.3: Put bearings together

Put the bearings onto the bolt with the flanges facing out.



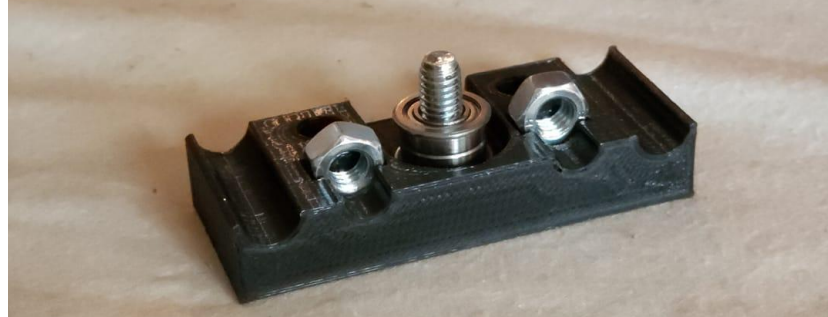
III. Instructional

Idler of X Axis

Step 15.2: Place nuts in idler half-piece

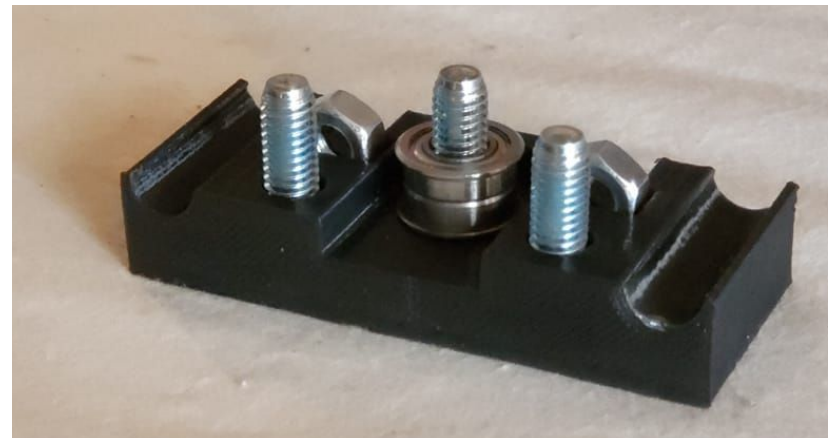
Put the nuts beside the bearing resting in the slot as shown.

These nuts will later be used to attach the X-Axis to the Y-Axis.



Step 15.3: Add [2] M6x18 bolts

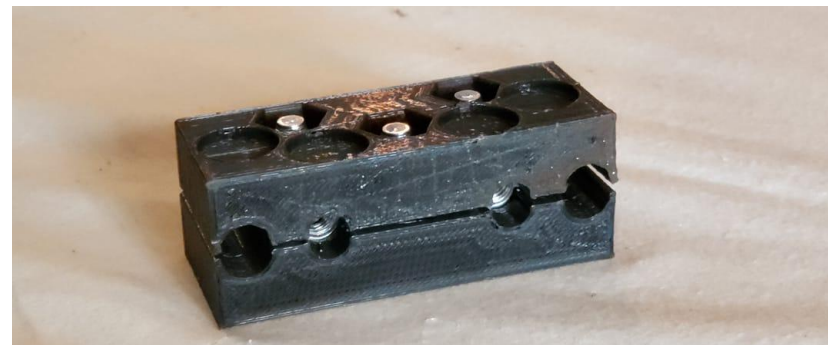
Add the 2 bolts through the idler half-piece as shown.



Step 15.4: Close up the idler

Close up the idler piece and tighten the bolts down fully with nuts and an electric drill on the lowest setting.

Make sure the idler piece is aligned with itself correctly, as pictured.

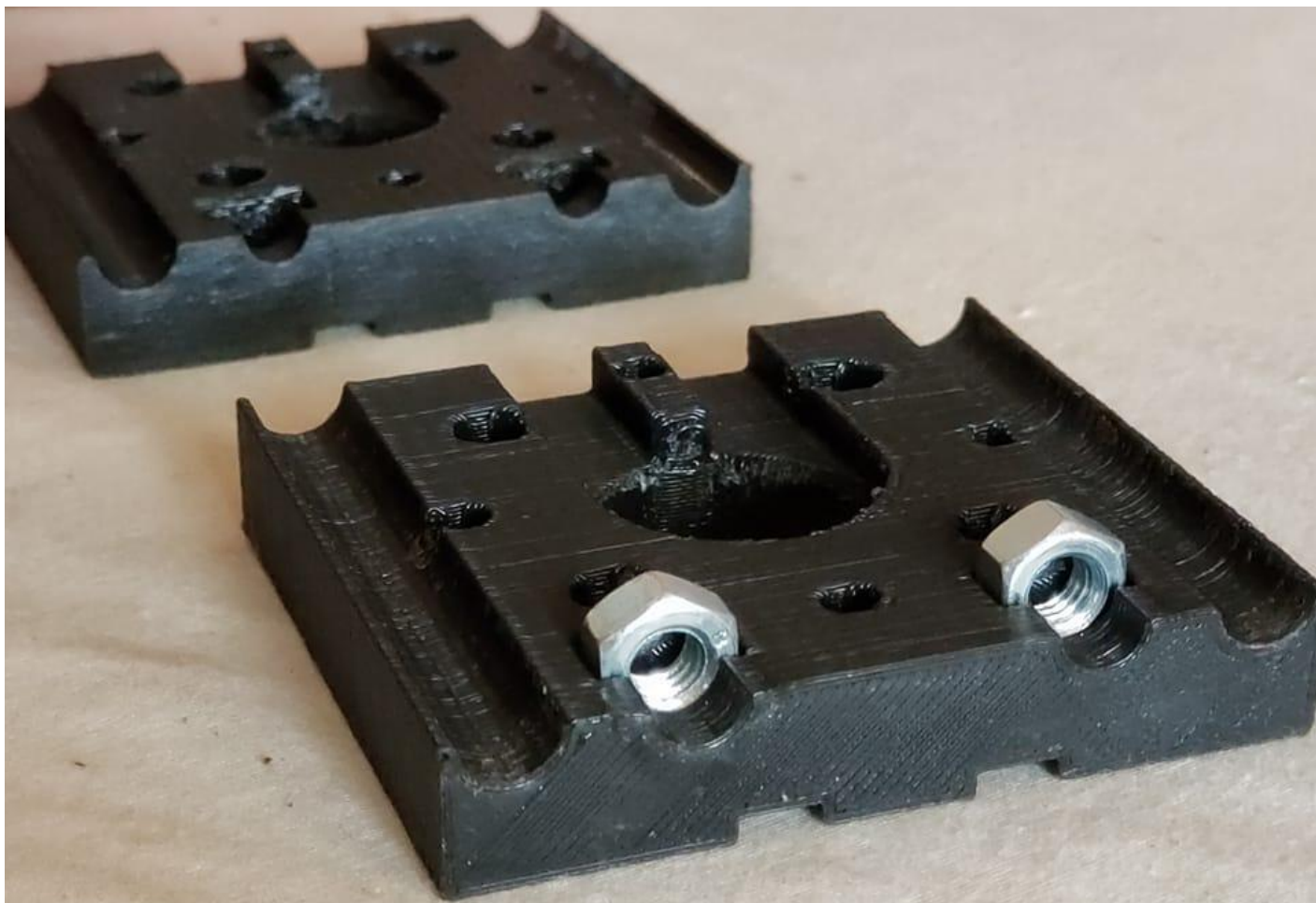


III. Instructional

X Axis

Step 16.1: Setup Motor Half-piece with two nuts

Set two nuts into the motor half-piece as shown.

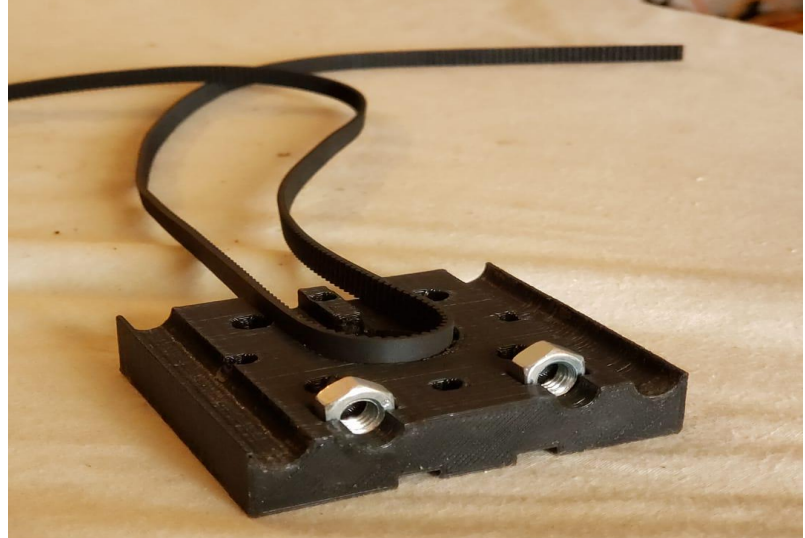


III. Instructional

X Axis

Step 16.2: Insert the belt

Place the belt in the round channel on the half-piece. Make sure the toothed side of the belt is facing inwards.



Step 16.3: Close up the motor

Close the motor half-pieces so they are properly aligned as shown.

Insert four nuts. No bolts yet.



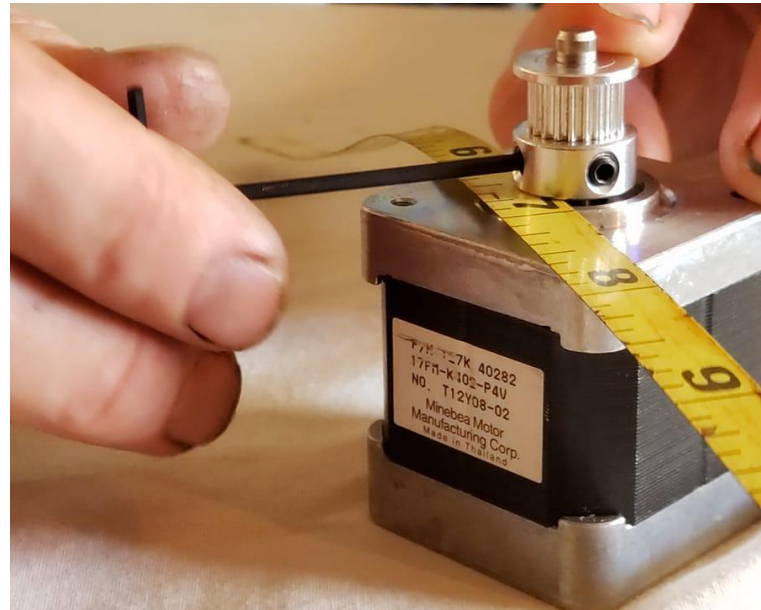
III. Instructional

X Axis

Step 17.1: Put pulley on motor

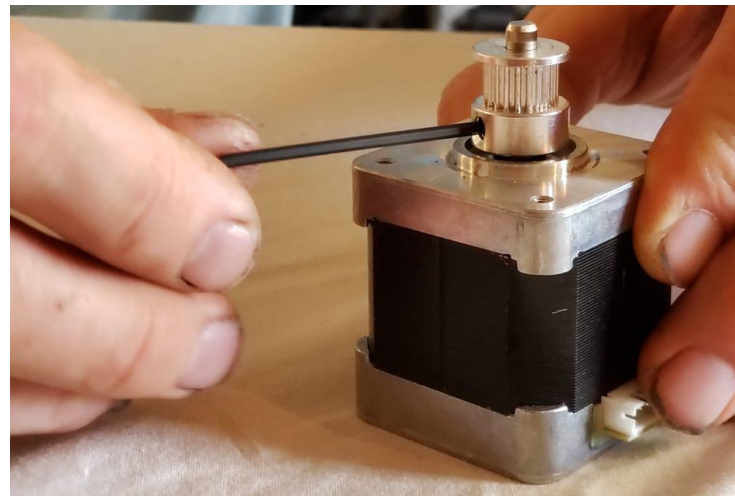
Place the pulley on the motor using a thin spacer (provided) that the pulley rests upon. Tighten the first set screw with an Allen wrench.

The spacer prevents the pulley from rubbing on the motor housing, and assists in aligning the pulley gear to the belt channel.



Step 17.2: Tighten pulley set screw onto motor shaft

With the first set screw tightened, you can remove the spacer. Then tighten the second set screw.



III. Instructional

X Axis

Step 17.3: Quality Control Check.

Flip the Allen wrench and tighten down the set screw as far as you can, pressing down on the wrench near its midpoint for greatest pressure. When you hear the set screw "pop.", you can be sure you've set it as tightly as possible.

The pop sound occurs when the small screw head slips along the motor shaft. It doesn't get any tighter than that. This tight fit ensures the belt will be pulled effectively by the motor shaft.

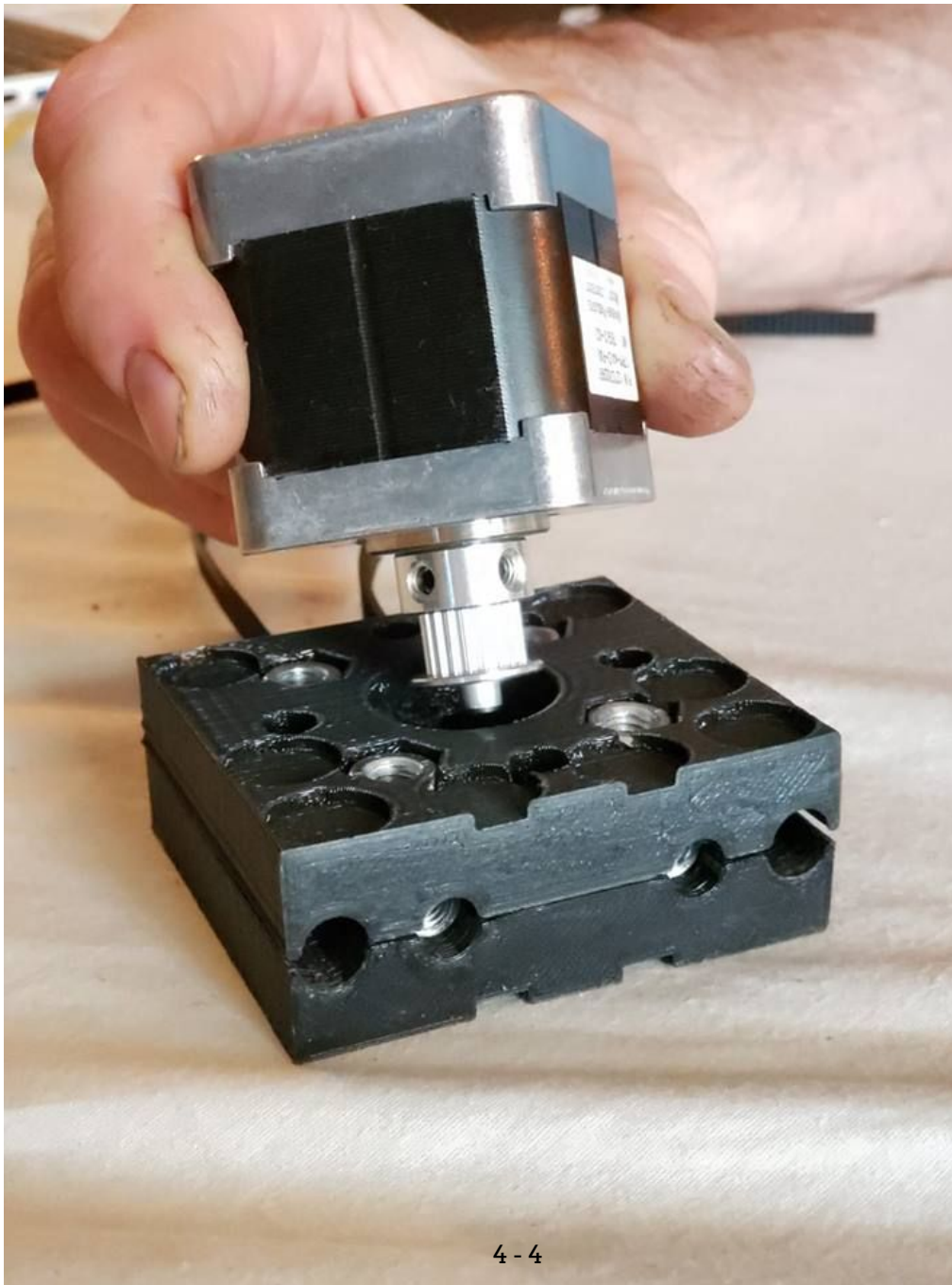


III. Instructional

X Axis

Step 17.4: Insert the motor

Place the motor on top as shown. Note the orientation of the motor -- with cable plug pointing towards the belt -- this is important.

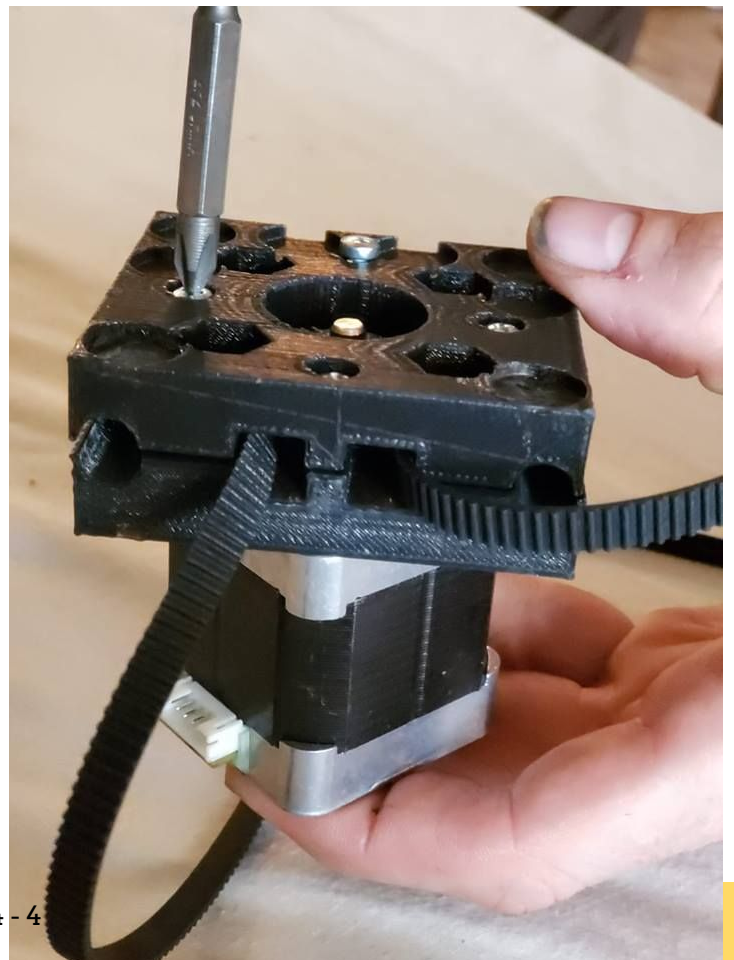


III. Instructional

X Axis

Step 17.5: Insert motor screws

Insert and loosely tighten four screws through the plastic pieces into the motor.

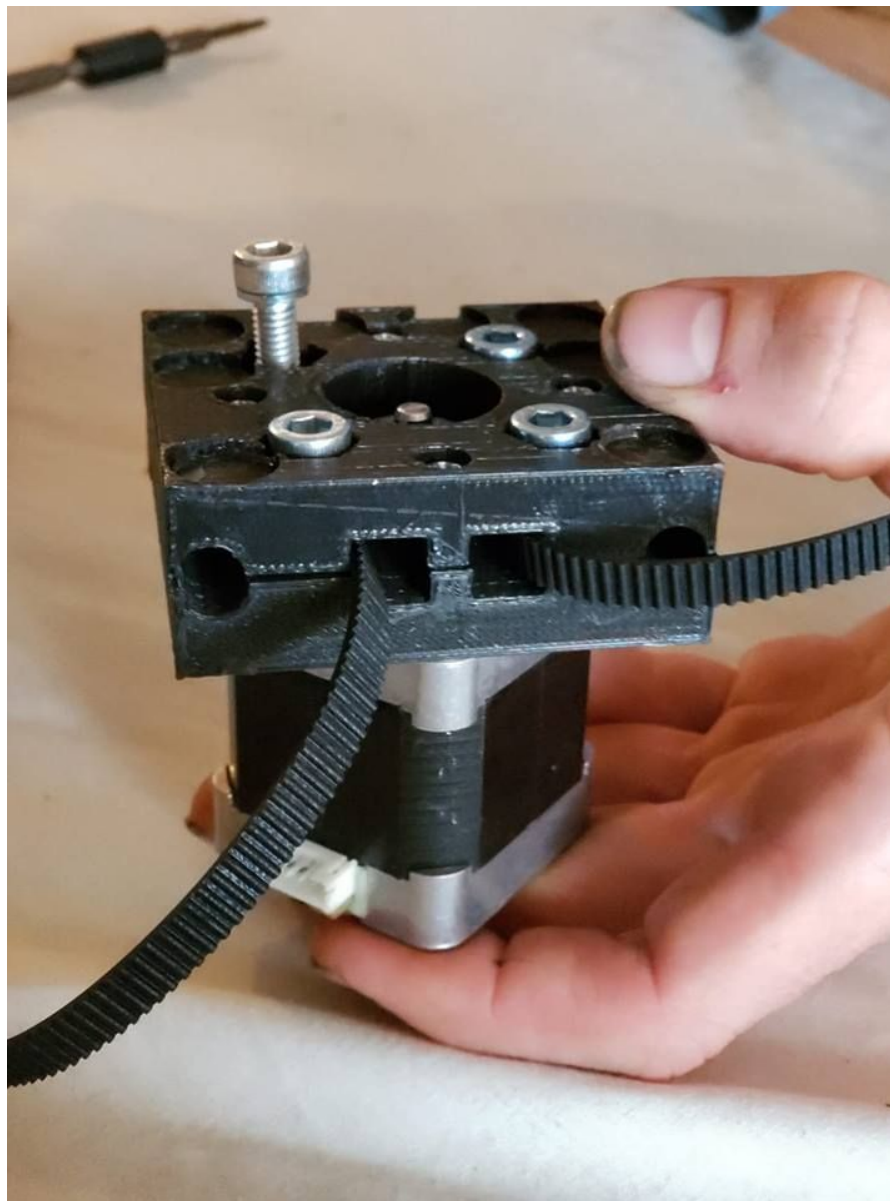


III. Instructional

X Axis

Step 17.6: Insert motor bolts

Insert and loosely tighten three M6x18 bolts and one M6x30 bolt as shown. Note the position of the longer bolt in relation to the belt hole.



III. Instructional

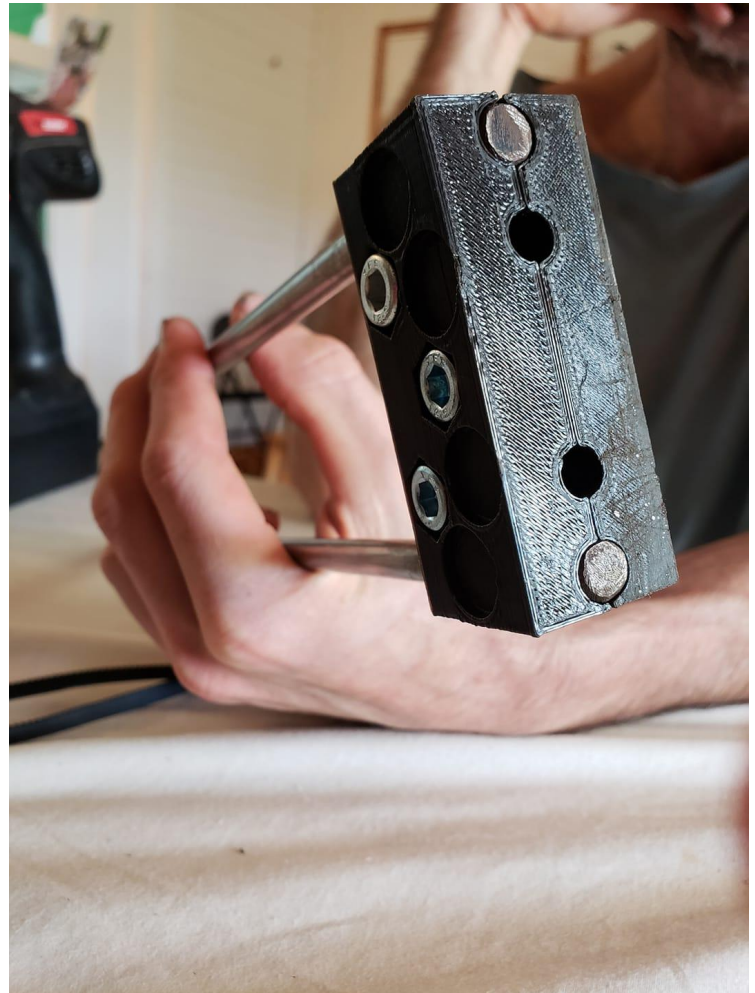
X Axis

Step 18.1 : Assemble Idler piece

Now we'll put it all together. Insert the rods into the completed idler piece.

Make sure the rods are flush against the bottom.

If it is difficult to fit the rods on, just loosen the bolts until the rods slide in. Then re-tighten.



III. Instructional

X Axis

Step 18.2: Put carriage assembly onto rods

Slide the carriage onto the rods.



Step 18.3: Put motor piece onto rods

Push motor piece onto the rods as shown. Note the orientation -- this is important.

If it is difficult to fit the rods on, just loosen the bolts until the rods slide in. Then re-tighten.



Step 18.4: **Quality Control Check**

Check the idler that the rods are still flush. (see image on previous page)

4 - 4
4 - 4

III. Instructional

X Axis

Step 18.5: Thread belt into carriage

You are going to tighten the belt through the motor piece, carriage, and idler. Before moving on, take a close look at the carriage piece. Notice it has two belt holes, one of which is larger than the other.

Insert one end of the belt through the larger belt hole on the carriage piece.



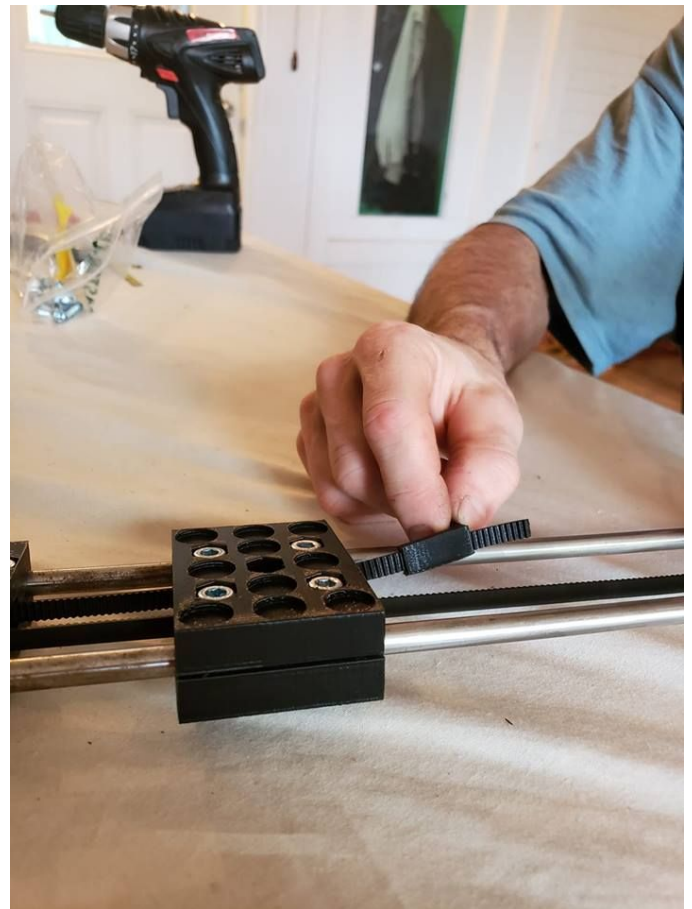
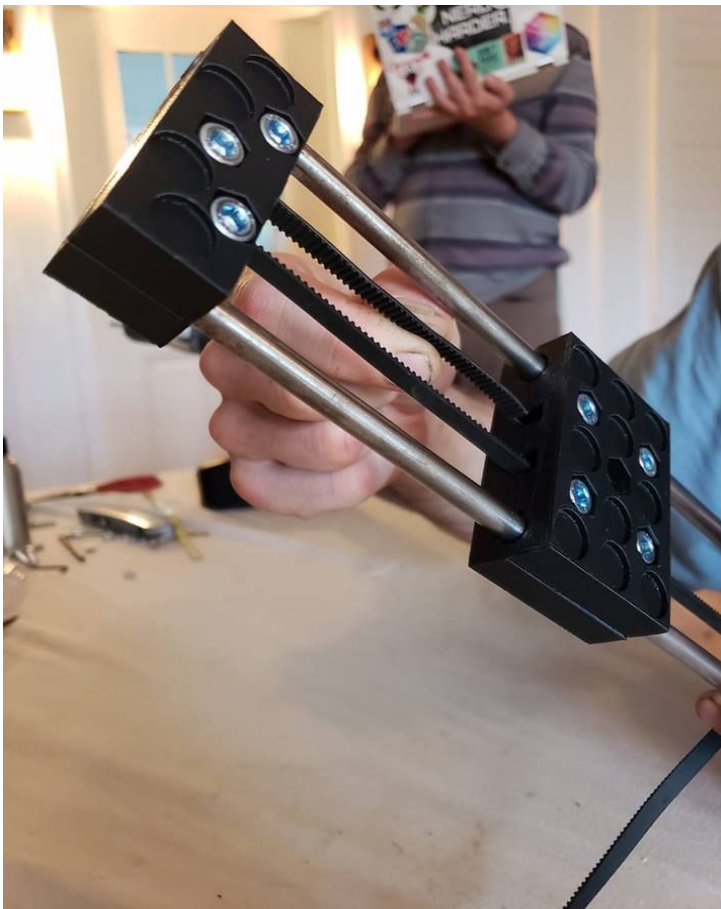
III. Instructional

X Axis

Step 19.1: Thread belt through one peg

Thread the belt end through one of the pegs. The “foot” of the peg should be turned away from the toothed side of the belt.

Quick check. *The foot should face away from the center of the axis assembly. The belt should be passing through the carriage before it gets put through the foot.*



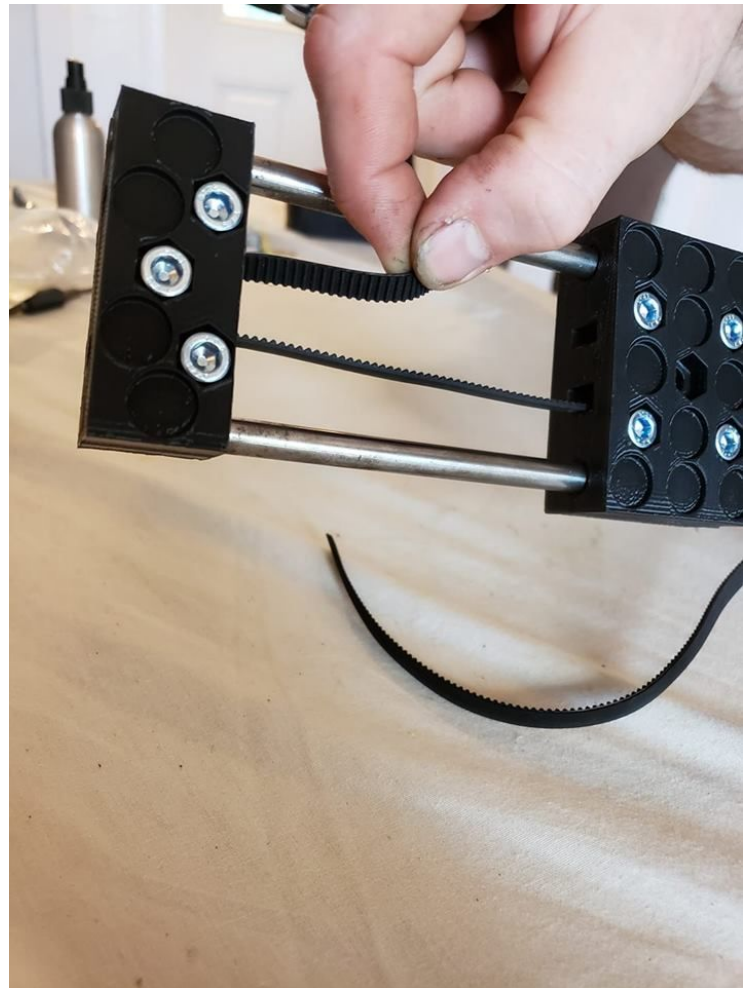
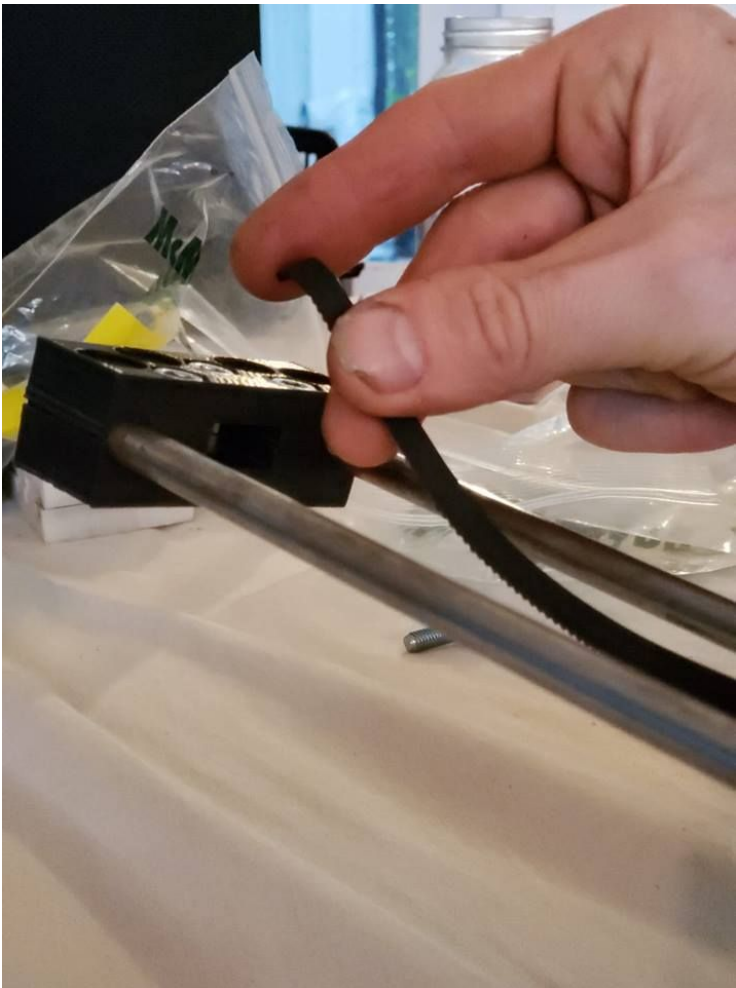
III. Instructional

X Axis

Step 18.5: Thread belt around idler bearing

Continue threading the belt around the bearing inside of the idler.

Tip: Give a slight curve in the right direction to your belt before sliding it inside the idler. It will tend to get around the bearing more easily.



III. Instructional

X Axis

Step 19.2: Tighten first peg into carriage

Insert the belt end into the belt hole, approximately $\frac{3}{4}$ ", then simultaneously insert the peg into the belt hole.

Then push the peg in tightly.



III. Instructional

X Axis

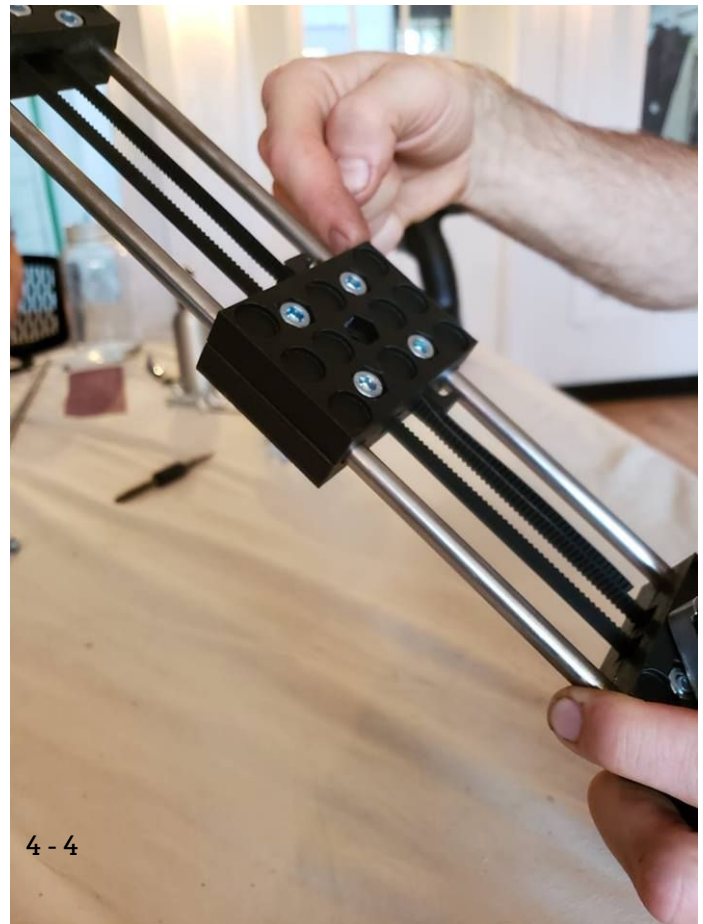
Step 19.3: Tighten second peg into carriage

Insert a second peg into the other side, directly across from the first peg.

Insert it loosely, then tighten the belt by hand by pulling it farther through the first peg just as much as can be done easily.

Leave this second peg loose, it will be tightened after fitting the Axis Assembly to the frame.

Test your build by ensuring the carriage can move smoothly along the rods.



4 - 4

III. Instructional

X Axis Only

Step 20.1: Gather an end stop holder piece and an end stop

Step 20.2 Slide the end stop into the end stop holder piece

Slide into the edge-crevices inside of the end stop holder piece.

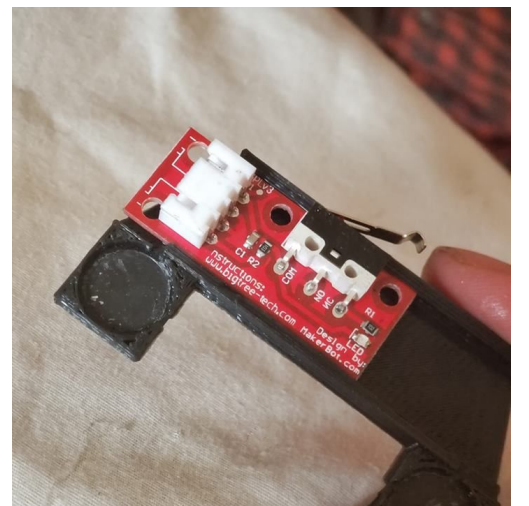
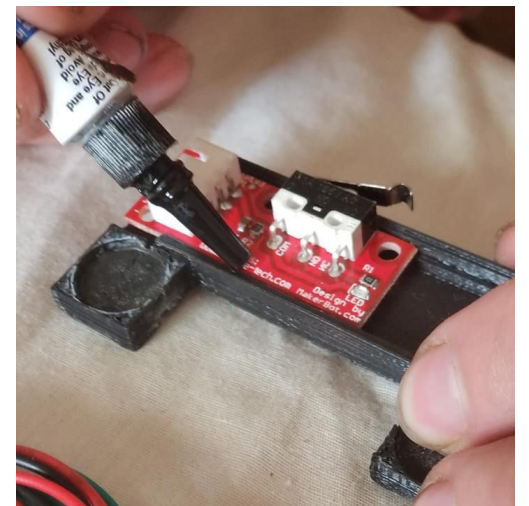
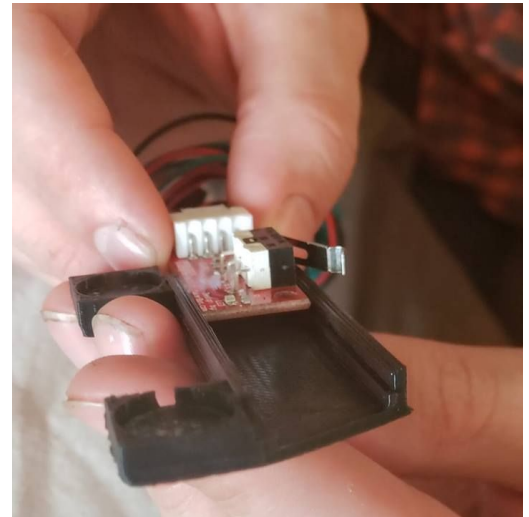
The switch arm should face away from the mounting legs and towards the arrow head that the end piece seems to be making as shown.

The bottom of the end stop should be flush with the bottom of the end stop holder piece.

Step 20.3: Super glue the end stop to the end stop holder piece

Dab super glue along the side edges of the crevice that the board slides through.

Do not put super glue on the metal switch of the end stop.



III. Instructional

X Axis Only

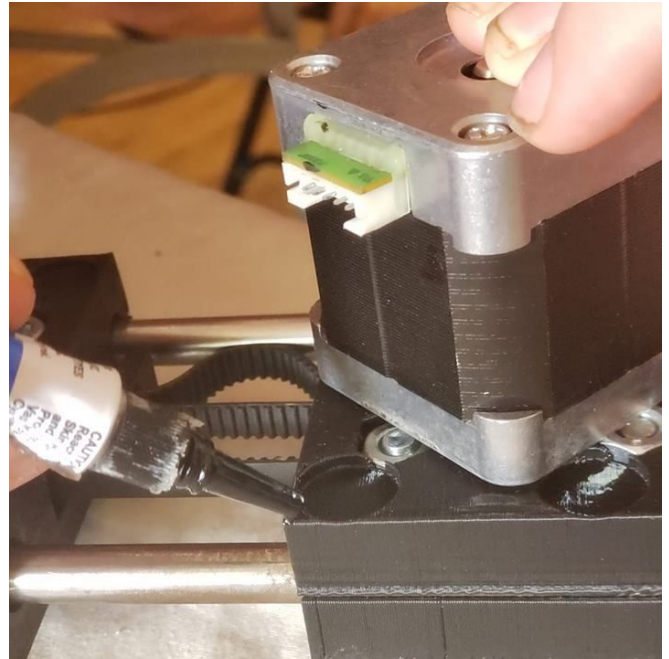
Step 21.1: Prep the Y-axis to have the end stop holder piece super glued to it
Dab super glue into the motor holder piece holes facing towards the belt. These will be the top left and bottom left holes if the motor piece is facing you.

Step 21.2: Super glue the end stop holder piece to the Y-axis motor holder piece

Press and hold the end stop holder piece to the motor holder piece.

Step 21.3: Add super glue along the edges of the end stop holder piece and the motor holder piece.

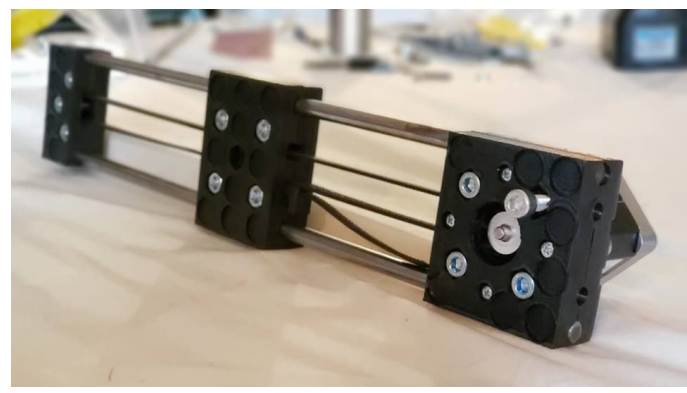
Adding this super glue here will make the connection a little stronger.



III. Instructional

X Axis

Step 22: Label the completed X-Axis



Step 23: Check that the X and Y Axes fit together

Align your completed X-Axis with the completed Y-Axes and make sure they will fit well together.



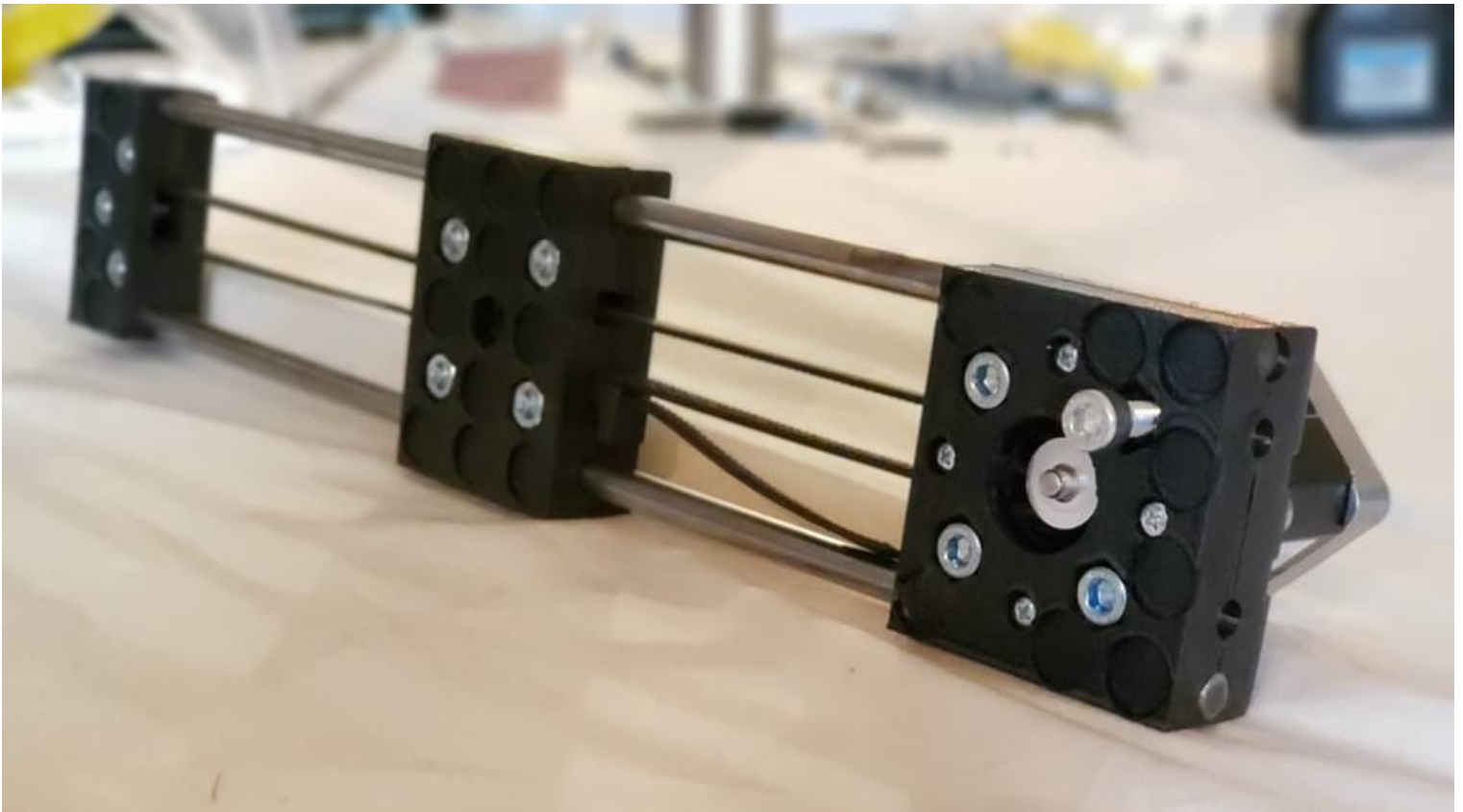
Step 24: Quality Control

- ☐ Carriage moves smoothly along axis rods
- ☐ Rods are flush with the idler bottom
- ☐ The idler and carriage bolt heads are on the same side
- ☐ Toothed side of belt is facing inwards
- ☐ Belt pegs are in the correct belt through-hole
- ☐ Belt peg holes line up

III. Instructional

Axes Quality Control

- ❑ Belt moves smoothly along axis rods
- ❑ Rods are flush with the idler bottom
- ❑ The idler and carriage bolt heads are on the same side
- ❑ Toothed side of belt is facing inwards
- ❑ Carriage moves smoothly along axis rods
- ❑ Belt peg holes line up
- ❑ Belt pegs are in the correct belt through-hole



I. Introduction

Bed Holder

The heated print bed will rest on the bed holder.

The bed holder is attached to the Z-front and Z-back axis by the carriages.

You'll be building two idlers just as in the Axis Assembly instructional except



I. Parts

Bed Holder

Step 20.1 Gather Parts for Bed Holder

See parts image on previous page.

Step 20.2 Put nuts on rods

Put two nuts on each bed rod.

These nuts will provide a flat surface for the bed to rest on during assembly, so the 3D printed objects can be printed well.

They are large nuts so that the wiring that goes beneath the heating bed can pass freely to the control panel. This will also minimize heat transfer from the bed to the plastic carriages of the axis that is holding the bed.



4 - 4

I. Parts

Bed Holder

Step 21.1 Prep Rod holders

Place three M6x18 bolts into the holes of the idler half piece.

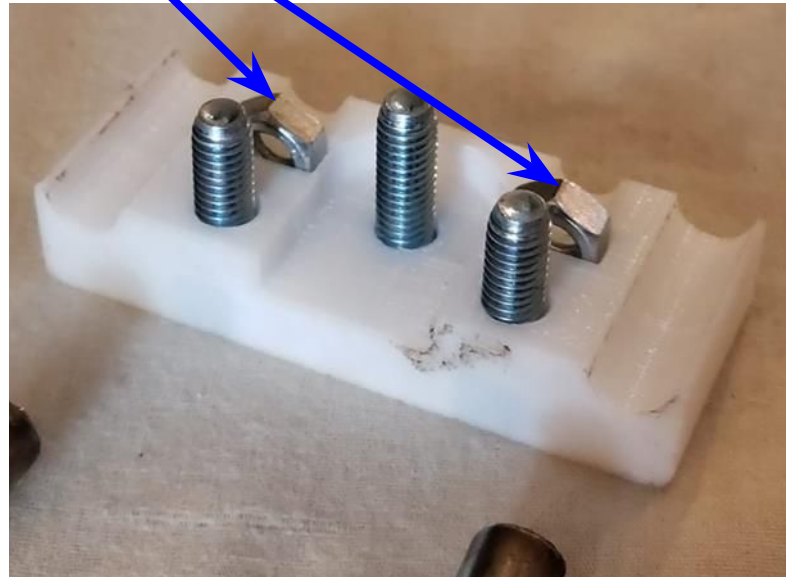


Nuts in nut catcher

Step 21.2 Place nuts inside catchers

Place nuts into gaps of idler as shown below.

Check that the nut catcher holes are free of loose plastic so a bolt can go in freely. If it's not clean, clean it with a razor.



I. Parts

Bed Holder

Step 21.3 Fasten idler pieces together

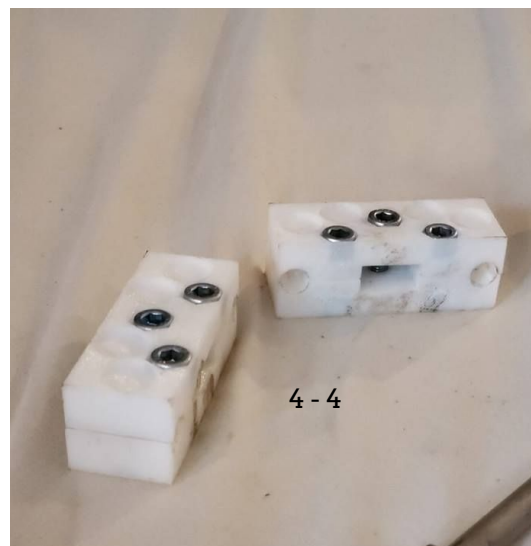
Put the idler half pieces together and place three nuts on top.

Turn the idler around so you can fasten it together with the nuts using an electric drill. Tighten them enough to keep the idler together but leave them slightly loose.



Step 22: Make another idler

Repeat steps 21.1 - 21.3 to make another idler exactly the same.



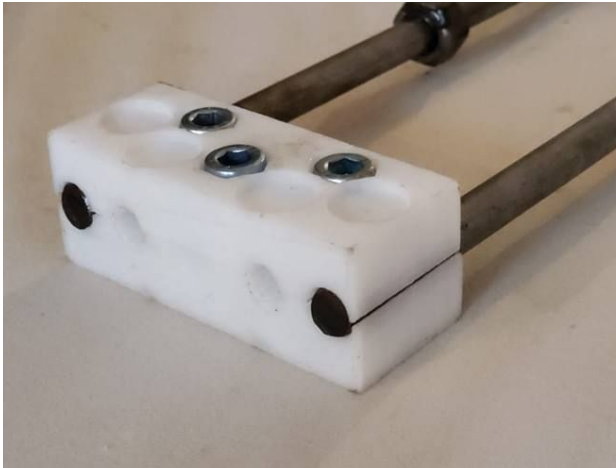
III. Instructional

Bed Holder

Step 22: Insert rods in the idler piece.

Slide the idler piece onto the rods so the rod ends are flush with the edge of the idler piece.

If needed, tighten the bolts with an electric drill but still leave them slightly loose. They'll be tightened down fully when this piece is assembled onto the frame.



A completed bed holder

IV. Completion Checklist

Bed Holder

- ☐ Rod ends are flush with the ends of the idler pieces
- ☐ Two nuts are present on each rod
- ☐ Bolts are tightened onto the rods
- ☐ Bolts are not over-tightened to the point of deforming the plastic
- ☐ Nuts inside nut catchers accept screws without cross-threading or spinning

V. Troubleshooting

Bed holder

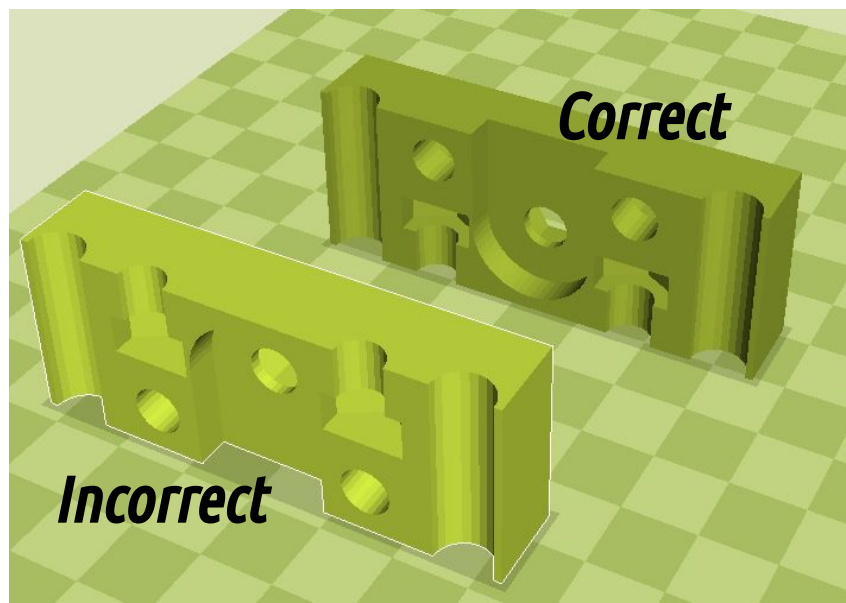
Screw not catching nut in idler

Imperfections in Idler piece:

Extra plastic may get printed by mistake. It shouldn't be there. You can use a box cutter to scrape it off the component.

Production Engineering:

When you print, wherever there is overhang, there will be roughness, so you want to print such that the curved part of idler piece does not have overhang.



VI. Design Considerations

Axis

Axis Design

The axes were designed so that a team of 5 or more people could build the axes all at the same time.

Further - the design is inherently scalable by lengthening, stacking, or paralleling multiple axes. Two motor printed pieces can also be used on one axis to double the torque. The axis parts themselves can be enlarged to 1" or even 3" rods for heavy duty machining. See [Universal Axis](#) on the wiki.

The end stops can be eliminated if special current-sensing drivers are used instead of the Pololu stepper drivers. Marlin - the 3D printer firmware - already has the capacity to handle the current-sensing drivers. These drivers plug into the same socket as the Pololu drivers.

Photo Repository

<https://drive.google.com/drive/u/1/folders/1uq-7aFNsjc0YFsg0ohCoaWsK-Pru94QI>